

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\HP Flash Gas KO Drum (155-D)Rev 0- 7%.pvdb

Date of Analysis : Sep 16,2026 2:03pm

PV Elite 28 U1, March 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1231.000 mm. ID	
Vessel Design Length, Tangent to Tangent	2441.50 mm.
Distance of Bottom Tangent above Grade	0.00 mm.
Specified Datum Line Distance	0.00 mm.
Internal Design Temperature	130 °C
Internal Design Pressure	40.998 bars
External Design Temperature	0 °C
Maximum Allowable Working Pressure	41 bars
Shop Test Pressure	56.107 bars
Field Test Pressure	53.297 bars
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-13.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 4	...	...	K03017	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Nozzle	SA-182 F316L	...	> 130	S31603	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F316L	...	> 130	S31603	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Leg Baseplate	SA-516 70	...	...	K02700	
Leg Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
BOTTOM HEAD	41.101	0.00	43.600	1.0000	No	No
SHELL-1	41.065	0.00	42.700	1.0000	N/A	No
SHELL-2	41.005	0.00	42.800	1.0000	N/A	No
BF-1	41.005	0.00	42.500	1.0000	N/A	No
BF-2	41.005	0.00	42.800	1.0000	N/A	No
TOP HEAD	41.005	0.00	43.700	1.0000	No	No

Liquid Level: 957.75 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.030

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	50.0	50.0	28.0	24.0	22.6	...	0.85	0.85



Cylinder	1475.0	1425.0	24.0	24.0	23.0	...	0.85	0.85
Cylinder	1927.0	452.0	24.0	24.0	23.0	...	0.85	0.85
Body Flg	2161.5	230.0	24.0	138.2	134.5	95.9	0.85	0.85
Body Flg	2391.5	230.0	24.0	138.2	130.0	77.7	0.85	0.85
Ellipse	2441.5	50.0	28.0	24.0	22.6	...	0.85	0.85

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Trunnion	WRC-107/537	0.149	n/a	Passed
Trunnion	WRC-107/537	0.150	n/a	Passed
Trunnion	WRC-107/537	0.150	n/a	Passed
Trunnion	WRC-107/537	0.150	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.150	n/a	Passed
Trunnion	WRC-107/537	0.150	n/a	Passed
Trunnion	WRC-107/537	0.149	n/a	Passed
Trunnion	WRC-107/537	0.149	n/a	Passed
Trunnion	WRC-107/537	0.148	n/a	Passed
Trunnion	WRC-107/537	0.148	n/a	Passed
Trunnion	WRC-107/537	0.147	n/a	Passed
Trunnion	WRC-107/537	0.146	n/a	Passed
Trunnion	WRC-107/537	0.146	n/a	Passed
Trunnion	WRC-107/537	0.145	n/a	Passed
Trunnion	WRC-107/537	0.144	n/a	Passed
Trunnion	WRC-107/537	0.143	n/a	Passed
Trunnion	WRC-107/537	0.142	n/a	Passed
Trunnion	WRC-107/537	0.141	n/a	Passed
Trunnion	WRC-107/537	0.140	n/a	Passed
Trunnion	WRC-107/537	0.139	n/a	Passed
Trunnion	WRC-107/537	0.138	n/a	Passed
Trunnion	WRC-107/537	0.137	n/a	Passed
Trunnion	WRC-107/537	0.136	n/a	Passed
Trunnion	WRC-107/537	0.135	n/a	Passed
Trunnion	WRC-107/537	0.133	n/a	Passed
Trunnion	WRC-107/537	0.132	n/a	Passed
Trunnion	WRC-107/537	0.131	n/a	Passed
Trunnion	WRC-107/537	0.129	n/a	Passed
Trunnion	WRC-107/537	0.128	n/a	Passed
Trunnion	WRC-107/537	0.126	n/a	Passed
Trunnion	WRC-107/537	0.125	n/a	Passed
Trunnion	WRC-107/537	0.123	n/a	Passed
Trunnion	WRC-107/537	0.122	n/a	Passed
Trunnion	WRC-107/537	0.120	n/a	Passed
Trunnion	WRC-107/537	0.118	n/a	Passed
Trunnion	WRC-107/537	0.117	n/a	Passed
Trunnion	WRC-107/537	0.115	n/a	Passed
Trunnion	WRC-107/537	0.113	n/a	Passed
Trunnion	WRC-107/537	0.114	n/a	Passed
Trunnion	WRC-107/537	0.115	n/a	Passed
Trunnion	WRC-107/537	0.117	n/a	Passed
Trunnion	WRC-107/537	0.119	n/a	Passed
Trunnion	WRC-107/537	0.121	n/a	Passed
Trunnion	WRC-107/537	0.123	n/a	Passed
Trunnion	WRC-107/537	0.124	n/a	Passed
Trunnion	WRC-107/537	0.126	n/a	Passed
Trunnion	WRC-107/537	0.128	n/a	Passed



Trunnion	WRC-107/537	0.130	n/a	Passed
Trunnion	WRC-107/537	0.131	n/a	Passed
Trunnion	WRC-107/537	0.133	n/a	Passed
Trunnion	WRC-107/537	0.135	n/a	Passed
Trunnion	WRC-107/537	0.136	n/a	Passed
Trunnion	WRC-107/537	0.138	n/a	Passed
Trunnion	WRC-107/537	0.140	n/a	Passed
Trunnion	WRC-107/537	0.141	n/a	Passed
Trunnion	WRC-107/537	0.143	n/a	Passed
Trunnion	WRC-107/537	0.145	n/a	Passed
Trunnion	WRC-107/537	0.146	n/a	Passed
Trunnion	WRC-107/537	0.148	n/a	Passed
Trunnion	WRC-107/537	0.149	n/a	Passed
Trunnion	WRC-107/537	0.151	n/a	Passed
Trunnion	WRC-107/537	0.152	n/a	Passed
Trunnion	WRC-107/537	0.154	n/a	Passed
Trunnion	WRC-107/537	0.155	n/a	Passed
Trunnion	WRC-107/537	0.157	n/a	Passed
Trunnion	WRC-107/537	0.158	n/a	Passed
Trunnion	WRC-107/537	0.160	n/a	Passed
Trunnion	WRC-107/537	0.162	n/a	Passed
Trunnion	WRC-107/537	0.163	n/a	Passed
Trunnion	WRC-107/537	0.165	n/a	Passed
Trunnion	WRC-107/537	0.166	n/a	Passed
Trunnion	WRC-107/537	0.168	n/a	Passed
Trunnion	WRC-107/537	0.170	n/a	Passed
Trunnion	WRC-107/537	0.172	n/a	Passed
Trunnion	WRC-107/537	0.173	n/a	Passed
Trunnion	WRC-107/537	0.175	n/a	Passed
Trunnion	WRC-107/537	0.177	n/a	Passed
Trunnion	WRC-107/537	0.179	n/a	Passed
Trunnion	WRC-107/537	0.181	n/a	Passed
Trunnion	WRC-107/537	0.181	n/a	Passed
LEGS	WRC-107/537	0.811	n/a	Passed
LEGS	WRC-107/537	0.521	n/a	Passed
N3-2"	WRC-107/537	0.717	n/a	Passed
N1-4"	WRC-297	0.812	Shell	Passed
N2-4"	WRC-107/537	0.717	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	222. Kg
Total Earthquake Shear on top of all Legs	1020. Kg
Total Wind Moment at top of all Legs	267. Kg-m.
Total Earthquake Moment at top of all Legs	1533. Kg-m.
Wind Load Case Test Moment on Support	165. Kg-m.
Earthquake Load Case Test Moment on Support	64. Kg-m.
Test Case Wind Shear on Support	111. Kg
Max. Wind Shear on one Leg (top & bottom)	88. Kg
Max. Earthq. Shear on one Leg (top & bottom)	406. Kg
Max. Wind Moment at base of one Leg	162. Kg-m.
Max. Earthquake Moment at base of one Leg	743. Kg-m.
Max. Vertical Load (Wt. + Wind) on one Leg	2196. Kg
Max. Vertical Load (Wt. + Eq.) on one Leg	3083. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	6888.3 kg.
Shop Test - Fabricated + water (full)	10276.7 kg.
Shipping - Fab. + rem. intls.+ shipping app.	6888.3 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	7340.5 kg.
Empty - Fab. + intls. + details + wghts.	7340.5 kg.
Operating - Empty + operating liquid (no ca)	8376.9 kg.
Field Test - Empty weight + test medium (full)	10663.8 kg.



Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3-2"	41	...	...	OK	5.80	OK	Passed
N1-4"	41	...	...	OK	6.26	OK	Passed
K1-2"	41	...	...	OK	5.80	OK	Passed
K2-2"	41	...	...	OK	5.80	OK	Passed
K3-2"	41	...	...	OK	5.80	OK	Passed
K4-2"	41	...	...	OK	5.80	OK	Passed
M1-24"	41	...	...	...		OK	Passed
N2-4"	41	...	...	OK	6.26	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 41 Nozzle : N2-4"
 Minimum MAWP Shells/Flanges : 41 Element : BF-1
 Minimum MAPnc Shells/Flanges : 43.3 Element : BF-1

Computed Vessel M.A.W.P. : 41 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	N3-2"	0.000	0.000	130.328
20	N1-4"	1250.000	90.000	188.099
20	K1-2"	900.000	90.000	130.328
20	K2-2"	500.000	90.000	130.328
20	K3-2"	650.000	270.000	130.328
20	K4-2"	300.000	270.000	130.328
20	M1-24"	900.000	180.000	1223.200
60	N2-4"	0.000	0.000	188.099

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	244.79	600
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	265.45	600
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	265.45	600
K3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	265.45	600
K4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	265.45	600
N1-4"	4.000 in	120	WNF	4.500	11.125	...	...	306.66	600
N2-4"	4.000 in	120	WNF	4.500	11.125	...	...	305.45	600
M1-24"	24.000 in	Actual	WNF	25.890	24.000	...	...	597.30	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3-2"	SA-106 B	19.500	10.000	...	...	...
K1-2"	SA-182 F316L	20.000	10.000	...	...	...
K2-2"	SA-182 F316L	20.000	10.000	...	...	...
K3-2"	SA-182 F316L	20.000	10.000	...	...	...
K4-2"	SA-182 F316L	20.000	10.000	...	...	...
N1-4"	SA-266 2	20.000	10.000	...	...	...
N2-4"	SA-266 2	19.500	10.000	...	...	...
M1-24"	SA-266 2	20.000	20.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N3-2"	...	0.0	220.00	0.00	BOTTOM HEAD
K1-2"	900.000	90.0	240.00	0.00	SHELL-1
K2-2"	500.000	90.0	240.00	0.00	SHELL-1
K3-2"	650.000	270.0	240.00	0.00	SHELL-1
K4-2"	300.000	270.0	240.00	0.00	SHELL-1
N1-4"	1250.000	90.0	280.00	0.00	SHELL-1
N2-4"	...	0.0	280.00	0.00	TOP HEAD
M1-24"	900.000	180.0	430.00	0.00	SHELL-1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 28.0mm. THK X 1231.0mm. ID X 50.0mm.	SA-516 70
2	WELD NECK FLANGE: 138.2mm. THK X 1530.0mm. OD	SA-266 4
2	INSULATION: 357mm X 80mm THK	
2	CLADDING: 357.750mm X 3.000mm THK	...
1	INSULATION: 1425mm X 80mm THK	
1	CLADDING: 1425.000mm X 3.000mm THK	...
4	LEGS, SHAPE: L150X150X15 1830.0mm	...
4	BASEPLATES: 350.0mm X 350.0mm X 26.0mm	SA-516 70
4	BASEPLATE BOLTS: 30.0mm	SA-307 A
4	LEG RE-PADS: 350.000mm X 250.000mm X 20.000mm	SA-516 70
1	INSULATION: 452mm X 80mm THK	
1	CLADDING: 452.000mm X 3.000mm THK	...
2	TRUNNIONS:	SA-516 70
2	TRUNNION PADS: 400.0mm OD X 168.3mm ID X 20.0mm THK	
2	INSULATION: 230mm X 80mm THK	
2	CLADDING: 230.000mm X 3.000mm THK	...
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	GASKET: 1346.2mm. OD X 1231.0mm. ID	...
32	BODY FLANGE BOLTS: 50.8mm. DIA	SA-193 B7
64	NUTS FOR BODY FLANGE BOLTS: 50.8mm. DIA	...
2	CYLINDRICAL SEGMENT 24.0mm. THK X 1538.7mm. X 1425.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm. THK X 789.8mm. X 1425.0mm.	SA-516 70
4	CYLINDRICAL SEGMENT 24.0mm. THK X 832.7mm. X 452.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 24.0mm. THK X 536.5mm. X 452.0mm.	SA-516 70
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
BF-1	[11]	C	-22	-25		0.948	24.000	1.00	No
BF-2	[11]	C	-22	-32		0.819	24.000	1.00	No
BOTTOM HEAD	[10]	D	-36	-47		0.797	24.000	0.85	No
BOTTOM HEAD	[7]	D	-32	-48		0.693	28.000	0.85	No
SHELL-1	[8]	D	-36	-46		0.813	24.000	0.85	No
SHELL-2	[8]	D	-36	-46		0.813	24.000	0.85	No
TOP HEAD	[10]	D	-36	-47		0.797	24.000	0.85	No
TOP HEAD	[7]	D	-32	-48		0.693	28.000	0.85	No
N3-2"	[1]	B	-13	-104		0.059	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N1-4"	[1]	B	-2	-13		0.811	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
M1-24"	[1]	B	-2	-13		0.811	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-29					
N2-4"	[1]	B	-2	-18		0.718	24.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Noz1. Bolting	[22]		-48			1.000	64.000	1.00	No
Bolting	[21]		-48						
Warmest MDMT:			-2	-13					
Required Minimum Design Metal Temperature						-12.0	°C		
Warmest Computed Minimum Design Metal Temperature						-13.0	°C		

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest) 40.998 bars
 Design Internal Temperature 130.0 °C
 Projection of Nozzle from Vessel Top 0 mm.
 Projection of Nozzle from Vessel Bottom 0 mm.
 Minimum Design Metal Temperature -12.0 °C
 Type of Construction Welded
 Special Service None
 Degree of Radiography RT-3
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(c)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE
 Load Case 13 IP+WE+EW
 Load Case 14 IP+WF+CW
 Load Case 15 IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	Km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		0	mm.
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm.
Distance Upwind of Crest	Lh	0	mm.
Distance from Crest to the Vessel	x	0	mm.
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm.
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		1.250	
Table Value Fa		1.300	
Table Value Fv		1.500	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		1.250	
Site Class		C	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		No	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
Overall M.A.W.P. + Static Head		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10



Element To Node	20	
Element Type	Elliptical	
Description	BOTTOM HEAD	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	1231	mm.
Element Thickness	24	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	28	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	40.998	bars
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	234.4	N./mm ²
UCS-66 Chart Curve Designation	D	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	-307.75	mm.
Height/Length of Liquid	357.75	mm.
Liquid Density	0.001028	kg./cm ³
Element From Node	10	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	-307.75	mm.
Height/Length of Insulation	357.75	mm.
Thickness of Insulation	80	mm.
Density	0.000352	kg./cm ³
Element From Node	10	
Detail Type	Lining	
Detail ID	SS 316L CLAD	
Dist. from "FROM" Node / Offset dist	-307.75	mm.
Height/Length of Lining	357.75	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm ³
Element From Node	10	
Detail Type	Nozzle	
Detail ID	N3-2"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	35.427	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	10	
Detail Type	Weight	
Detail ID	Anti Swirl Baff	
Dist. from "FROM" Node / Offset dist	-307	mm.



Miscellaneous Weight	0	Kgf
Offset from Element Centerline	0	mm.

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	SHELL-1
Distance "FROM" to "TO"	1425 mm.
Inside Diameter	1231 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	24 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	40.998 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.33
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	600 mm.
Liquid Density	0.001028 kg./cm3

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1425 mm.
Thickness of Insulation	80 mm.
Density	0.000352 kg./cm3

Element From Node	20
Detail Type	Lining
Detail ID	Lining
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	1425 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3

Element From Node	20
Detail Type	Nozzle
Detail ID	N1-4"
Dist. from "FROM" Node / Offset dist	1200 mm.
Nozzle Diameter	4 in.
Nozzle Schedule	120
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	32.759 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	20
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	850 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.837	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K2-2"	
Dist. from "FROM" Node / Offset dist	450	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.837	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K3-2"	
Dist. from "FROM" Node / Offset dist	600	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.837	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K4-2"	
Dist. from "FROM" Node / Offset dist	250	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.837	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	850	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	904.81	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	20	
Detail Type	Leg	
Detail ID	LEGS	
Dist. from "FROM" Node / Offset dist	250	mm.
Diameter at Leg Centerline	1425.9	mm.
Leg Orientation	3	
Number of Legs	4	
Section Identifier	L150X150X15	
Length of Legs	1830	mm.

Element From Node	20	
Detail Type	Weight	
Detail ID	Davit Weight-M1	



Dist. from "FROM" Node / Offset dist	850 mm.
Miscellaneous Weight	50 Kg
Offset from Element Centerline	1275 mm.

Element From Node	20
Detail Type	Weight
Detail ID	Clip Weight
Dist. from "FROM" Node / Offset dist	710 mm.
Miscellaneous Weight	814 Kg
Offset from Element Centerline	0 mm.

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	SHELL-2
Distance "FROM" to "TO"	452 mm.
Inside Diameter	1231 mm.
Element Thickness	24 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	24 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	40.998 bars
Design Temperature Internal Pressure	130 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.33
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	30
Detail Type	Insulation
Detail ID	Insulation-3
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	452 mm.
Thickness of Insulation	80 mm.
Density	0.000352 kg./cm3

Element From Node	30
Detail Type	Lining
Detail ID	Lining
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	452 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3

Element From Node	30
Detail Type	Weight
Detail ID	Demisting Pad
Dist. from "FROM" Node / Offset dist	276 mm.
Miscellaneous Weight	26 Kg
Offset from Element Centerline	0 mm.

Element From Node	30
Detail Type	Weight
Detail ID	Demisting Suppo
Dist. from "FROM" Node / Offset dist	276 mm.
Miscellaneous Weight	32 Kg
Offset from Element Centerline	0 mm.

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Flange



Description	BF-1	
Distance "FROM" to "TO"	230	mm.
Flange Inside Diameter	1231	mm.
Element Thickness	138.2	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	24	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	40.998	bars
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-266 4	
Allowable Stress, Ambient	138	N./mm2
Allowable Stress, Operating	138	N./mm2
Allowable Stress, Hydrotest	225	N./mm2
Material Density	0.00775	kg./cm3
P Number Thickness	32.004	mm.
Yield Stress, Operating	222.2	N./mm2
UCS-66 Chart Curve Designation	C	
External Pressure Chart Name	CS-2	
UNS Number	K03017	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	230	mm.
Thickness of Insulation	80	mm.
Density	0.000352	kg./cm3
Element From Node	40	
Detail Type	Lining	
Detail ID	Lining	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	230	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm3

Element#	5/6	
Element From Node	50	
Element To Node	60	
Element Type	Flange	
Description	BF-2	
Distance "FROM" to "TO"	230	mm.
Flange Inside Diameter	1231	mm.
Element Thickness	138.2	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	24	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	40.998	bars
Design Temperature Internal Pressure	130	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.33	
Material Name	SA-266 4	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	230	mm.
Thickness of Insulation	80	mm.



Density 0.000352 kg./cm3

Element From Node 50
 Detail Type Lining
 Detail ID Lining
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Lining 230 mm.
 Thickness of Lining 3 mm.
 Density 0.008 kg./cm3

Element# 6/6
 Element From Node 60
 Element To Node 70
 Element Type Elliptical
 Description TOP HEAD
 Distance "FROM" to "TO" 50 mm.
 Inside Diameter 1231 mm.
 Element Thickness 24 mm.
 Internal Corrosion Allowance 1 mm.
 Nominal Thickness 28 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 40.998 bars
 Design Temperature Internal Pressure 130 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.33
 Material Name SA-516 70
 Allowable Stress, Ambient 138 N./mm2
 Allowable Stress, Operating 138 N./mm2
 Allowable Stress, Hydrotest 234 N./mm2
 Material Density 0.00775 kg./cm3
 P Number Thickness 32.004 mm.
 Yield Stress, Operating 234.4 N./mm2
 UCS-66 Chart Curve Designation D
 External Pressure Chart Name CS-2
 UNS Number K02700
 Product Form Plate
 Efficiency, Longitudinal Seam 0.85
 Efficiency, Circumferential Seam 0.85
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 60
 Detail Type Insulation
 Detail ID Insulation
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 357.75 mm.
 Thickness of Insulation 80 mm.
 Density 0.000352 kg./cm3

Element From Node 60
 Detail Type Lining
 Detail ID Lining
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Lining 357.75 mm.
 Thickness of Lining 3 mm.
 Density 0.008 kg./cm3

Element From Node 60
 Detail Type Nozzle
 Detail ID N2-4"
 Dist. from "FROM" Node / Offset dist 0 mm.
 Nozzle Diameter 4 in.
 Nozzle Schedule 120
 Nozzle Class 600
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 33.371 Kgf

PV Elite 28 U1 Licensee: SL User
FileName : HP Flash Gas KO Drum (155-D)_Rev 0- 7% Page 19 of 141
Input Echo: Step: 1 2:03pm Sep 16,2026



HEXAGON

Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
BOTTOM HEAD		...	50	...	50
SHELL-1		...	1475	...	1425
SHELL-2		...	1927	...	452
BF-1		...	2161.5	...	230
BF-2		...	2391.5	...	230
TOP HEAD		...	2441.5	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

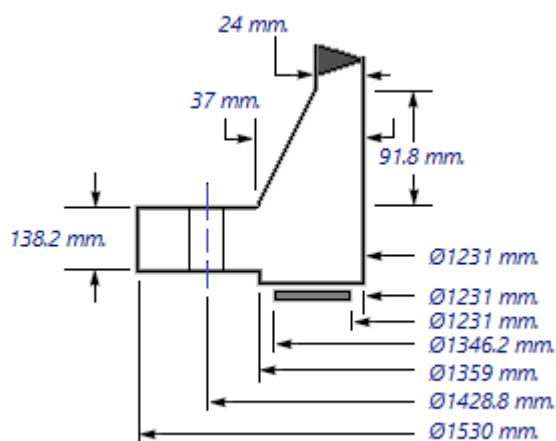
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Flange Input Data Values

Description: FLANGE :

BF-1

Flange Type	Integral Weld Neck		
Design Pressure	P	41.00	bars
Design Temperature		130	°C
Internal Corrosion Allowance	ci	1.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1231.000	mm.
Flange Outside Diameter	A	1530.000	mm.
Flange Thickness	t	138.2000	mm.
Thickness of Hub at Small End	go	24.0000	mm.
Thickness of Hub at Large End	g1	37.0000	mm.
Length of Hub	h	91.8000	mm.
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm2
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm2
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm2
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm2
Diameter of Bolt Circle	C	1428.750	mm.
Nominal Bolt Diameter	a	50.8000	mm.
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1359.000	mm.
Flange Face Inside Diameter	Fid	1231.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1346.200	mm.
Gasket Inside Diameter	Gi	1231.000	mm.
Gasket Factor	m	2.5000	
Gasket Design Seating Stress	y	68.95	N./mm2
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.
Axial Force on Flange		1267.59	Kgf
Bending Moment on Flange		73.22	Kg-m.



Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (41 \cdot (1231/2 + 1)) / (138 \cdot 0.85 - 0.6 \cdot 41) + Ca$$

$$= 23.0140 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 23) / (616 + 0.6 \cdot 23)$$

$$= 42.801 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (616 + 0.6 \cdot 24)$$

$$= 44.690 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	1233.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	36.000 mm.
Corroded Small Hub,	g0Cor = go-ci	23.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	61.875 mm.
Gasket Contact Width,	N = (Go - Gi)/2	56.600 mm.
Basic Gasket Width,	bo = N/2	28.300 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	13.405 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	1319.389 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1319^2 \cdot 41$$

$$= 571695.875 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 13.4 \cdot \pi \cdot 1319 \cdot 2.5 \cdot 41$$

$$= 116172.188 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1233^2 \cdot 41$$

$$= 499281.406 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 571696 - 499281$$

$$= 72414.461 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(571696 + 116172 + 0, 0)$$

$$= 687868.062 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 13.4 \cdot \pi \cdot 1319 + 0 \cdot 0 \cdot 0$$

$$= 390668.500 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((687868 + 1268 + 4 \cdot 73.2/1319) / 172, 390668/172)$$

$$= 393.054 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 50.8 + 6 \cdot 138/(2.5 + 0.5)$$

$$= 378.000 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1429 \cdot \sin(\pi / 32)$$

= 140.042 mm.

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max\left(\sqrt{\frac{Bs}{(2a + t)}}, 1\right)$$

$$= \max\left(\sqrt{\frac{140}{(2 \cdot 50.8 + 138)}}, 1\right)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	393.054	547.509	
Radial Distance between Hub and Bolts:	63.500	61.875	
Radial Distance between Bolts and Edge:	50.800	50.625	
Circ. Spacing between the Bolts:	107.950	140.042	378.000

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 548 \cdot 172 / (68.9 \cdot \pi(1346 + 1231))$$

$$= 16.869 \text{ mm.}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa (Am + Ab) / 2$$

$$= 172 (393 + 548) / 2$$

$$= 824802.19 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 687868 - 571696$$

$$= 116172.17 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1429 - 1319) / 2$$

$$= 54.6804 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1429 - 1233) / 2 + 54.7)$$

$$= 76.2777 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1429 - 1233 - 36) / 2$$

$$= 79.8750 \text{ mm.}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 148, Bb = 138, Cc = 91.8 \text{ and } Ddg = 29.5 \text{ mm.}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 148 \cdot 138^3 \cdot (1/3 - 0.210 \cdot (138/148) \cdot (1 - 1/12 \cdot (138/148)^4))$$

$$= 5884.039 \text{ cm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.8 \cdot 29.5^3 \cdot (1/3 - 0.105 \cdot (29.5/91.8) \cdot (1 - 1/192 \cdot (29.5/91.8)^4))$$

$$= 70.606 \text{ cm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 3591 + 43.1$$

= 5954.645 cm⁴

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 148 \cdot 138^3 \left(\frac{1}{3} - 0.21 \left(\frac{138}{148} \right) \left(1 - \frac{1}{12} \left(\frac{138}{5.85} \right)^4 \right) \right)$$

$$= 5884.039 \text{ cm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip, term1, Ip, term2)$$

$$= \max(5955, 5884)$$

$$= 5954.6 \text{ cm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.78 \cdot 23^2 \cdot 168 \cdot 1233 / 0.29$$

$$= 5910.502 \text{ cm}^4$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	499281.	79.8750	1.0000	39881.
Face Pressure, Mt	72414.	76.2777	1.0000	5524.
Gasket Load, Mg	116172.	54.6804	1.0000	6352.
Gasket Seating, Matm	824802.	54.6804	1.0000	45101.
Moment Load, Moe	...	...	...	115.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 73.2 \left[\frac{5911}{(0.384 \cdot 5955 + 5911)} \right] \cdot \left(\frac{79.9}{(1429 - 2 \cdot 79.9)} \right) + 1268 \cdot 79.9$$

$$= 114.544 \text{ Kg-m.}$$

Total Moment for Operation, Mo 51872. Kg-m.
 Total Moment for Gasket seating, Mg 45101. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 168.401 mm.
 Hub Ratio, h/h0 = HL / H0 0.545
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.565

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.836
 Factor V per 4.16.5 0.289
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1530 / 1233$$

$$= 1.241$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.24^2 \cdot \log_{10}(1.24))) / (1.24^2 - 1) / (1.24 - 1)$$

$$= 9.121$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.24^2(1 + 8.55246 \cdot \log_{10}(1.24)) - 1) / ((1.0472 + 1.9448 \cdot 1.24^2) \cdot (1.24 - 1))$$

$$= 1.822$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.24^2(1 + 8.55246 \cdot \log_{10}(1.24)) - 1) / (1.36136(1.24^2 - 1) \cdot (1.24 - 1))$$

$$= 10.023$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.24^2 + 1) / (1.24^2 - 1)$$

$$= 4.705$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 10 \cdot 23^2 \cdot 168 / 0.29$$

$$= 3089074.250 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.84 / 168$$

$$= 0.005 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (138 \cdot 0.005 + 1) / 1.82 + 138^3 / 3089074$$

$$= 1.780$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 51872 / 1233) / (1.78 \cdot 36^2)$$

$$= 178.88 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 45101 / 1233) / (1.78 \cdot 36^2)$$

$$= 155.53 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.91 \cdot 51872 / 1233) / (1.78 \cdot 138^2)$$

$$= 23.24 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.91 \cdot 45101 / 1233) / (1.78 \cdot 138^2)$$

$$= 20.20 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (9.12 \cdot 51872 / (138^2 \cdot 1233)) - 4.71 \cdot 23.2$$

$$= 87.68 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (9.12 \cdot 45101 / (138^2 \cdot 1233)) - 4.71 \cdot 20.2$$

$$= 76.24 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (179 + \max(23.2, 87.7)) / 2$$

$$= 133.28 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (156 + \max(20.2, 76.2)) / 2$$

$$= 115.89 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= (Wm_1 + Fa + 4Me/G) / Ab$$

$$= 689358 / 548$$

$$= 123.48 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm_2 / Ab)$$

$$= (390668 / 548)$$

$$= 69.98 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	178.88	207.00	155.53	207.00
Radial Flange	23.24	138.00	20.20	138.00
Tangential Flange	87.68	138.00	76.24	138.00
Maximum Average	133.28	138.00	115.89	138.00
Bolting	123.48	172.00	69.98	172.00

Minimum Required Flange Thickness 134.468 mm.

Estimated M.A.W.P. (Operating) 42.5 bars

Estimated Finished Weight of Flange at given Thk. 780.5 kg.

Estimated Unfinished Weight of Forging at given Thk 1155.7 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 45101 / 1 \cdot 9807 \cdot 0.29 / (1.7 \cdot 201334 \cdot 23^2 \cdot 168 \cdot 0.3)$
 $= 0.727$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 51872 / 1 \cdot 9807 \cdot 0.29 / (1.7 \cdot 194922 \cdot 23^2 \cdot 168 \cdot 0.3)$
 $= 0.864$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 45101 / 1 \cdot 9807 \cdot 0.29 / (1.78 \cdot 201334 \cdot 23^2 \cdot 168 \cdot 0.3)$
 $= 0.696$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 51872 / 1 \cdot 9807 \cdot 0.29 / (1.78 \cdot 194922 \cdot 23^2 \cdot 168 \cdot 0.3)$
 $= 0.827$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.95, Temperature Reduction per Fig. UCS 66.1 = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -22 °C

Min Metal Temp. at Required thickness (UCS 66.1) -25 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	177.19	207.00	153.47	207.00
Radial Flange	24.12	138.00	20.89	138.00
Tangential Flange	84.50	138.00	73.19	138.00
Maximum Average	130.84	138.00	113.33	138.00
Bolting	123.62	172.00	70.58	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 132.283 mm.

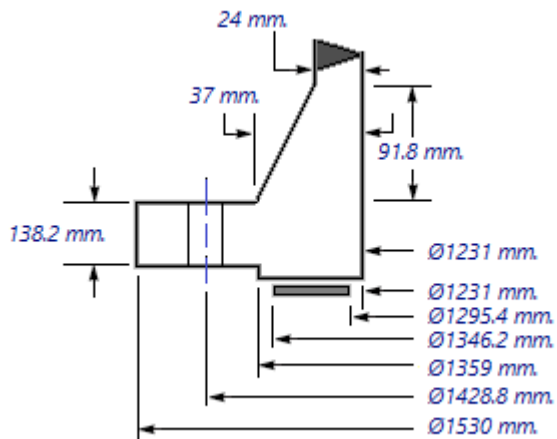
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Flange Input Data Values

Description: FLANGE :

BF-2

Flange Type	Integral Weld Neck		
Design Pressure	P	41.00	bars
Design Temperature		130	°C
Internal Corrosion Allowance	ci	1.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1231.000	mm.
Flange Outside Diameter	A	1530.000	mm.
Flange Thickness	t	138.2000	mm.
Thickness of Hub at Small End	go	24.0000	mm.
Thickness of Hub at Large End	gl	37.0000	mm.
Length of Hub	h	91.8000	mm.
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm2
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm2
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm2
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm2
Diameter of Bolt Circle	C	1428.750	mm.
Nominal Bolt Diameter	a	50.8000	mm.
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	1359.000	mm.
Flange Face Inside Diameter	Fid	1231.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1346.200	mm.
Gasket Inside Diameter	Gi	1295.400	mm.
Gasket Factor	m	2.5000	
Gasket Design Seating Stress	y	68.95	N./mm2
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.
Axial Force on Flange		486.84	Kgf
Bending Moment on Flange		33.51	Kg-m.



Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (41 \cdot (1231/2 + 1)) / (138 \cdot 0.85 - 0.6 \cdot 41) + Ca$$

$$= 23.0140 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 23) / (616 + 0.6 \cdot 23)$$

$$= 42.801 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (616 + 0.6 \cdot 24)$$

$$= 44.690 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	1233.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	36.000 mm.
Corroded Small Hub,	g0Cor = go-ci	23.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	61.875 mm.
Gasket Contact Width,	N = (Go - Gi)/2	25.400 mm.
Basic Gasket Width,	bo = N/2	12.700 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	8.980 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	1328.240 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1328^2 \cdot 41$$

$$= 579391.312 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.98 \cdot \pi \cdot 1328 \cdot 2.5 \cdot 41$$

$$= 78345.539 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1233^2 \cdot 41$$

$$= 499281.406 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 579391 - 499281$$

$$= 80109.898 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(579391 + 78346 + 0, 0)$$

$$= 657736.875 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 8.98 \cdot \pi \cdot 1328 + 0 \cdot 0 \cdot 0$$

$$= 263463.531 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((657737 + 487 + 4 \cdot 33.5/1328) / 172, 263464/172)$$

$$= 375.360 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 50.8 + 6 \cdot 138/(2.5 + 0.5)$$

$$= 378.000 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1429 \cdot \sin(\pi / 32)$$

= 140.042 mm.

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max\left(\sqrt{\frac{Bs}{(2a + t)}}, 1\right)$$

$$= \max\left(\sqrt{\frac{140}{(2 \cdot 50.8 + 138)}}, 1\right)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	375.360	547.509	
Radial Distance between Hub and Bolts:	63.500	61.875	
Radial Distance between Bolts and Edge:	50.800	50.625	
Circ. Spacing between the Bolts:	107.950	140.042	378.000

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 548 \cdot 172 / (68.9 \cdot \pi(1346 + 1295))$$

$$= 16.457 \text{ mm.}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa (Am + Ab) / 2$$

$$= 172 (375 + 548) / 2$$

$$= 809285.69 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 657737 - 579391$$

$$= 78345.56 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1429 - 1328) / 2$$

$$= 50.2553 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((1429 - 1233) / 2 + 50.3)$$

$$= 74.0651 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1429 - 1233 - 36) / 2$$

$$= 79.8750 \text{ mm.}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 148, Bb = 138, Cc = 91.8 \text{ and } Ddg = 29.5 \text{ mm.}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 148 \cdot 138^3 \cdot (1/3 - 0.210 \cdot (138/148) \cdot (1 - 1/12 \cdot (138/148)^4))$$

$$= 5884.039 \text{ cm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 91.8 \cdot 29.5^3 \cdot (1/3 - 0.105 \cdot (29.5/91.8) \cdot (1 - 1/192 \cdot (29.5/91.8)^4))$$

$$= 70.606 \text{ cm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 3591 + 43.1$$

= 5954.645 cm⁴

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 148 \cdot 138^3 \left(\frac{1}{3} - 0.21 \left(\frac{138}{148} \right) \left(1 - \frac{1}{12} \left(\frac{138}{5.85} \right)^4 \right) \right)$$

$$= 5884.039 \text{ cm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip, term1, Ip, term2)$$

$$= \max(5955, 5884)$$

$$= 5954.6 \text{ cm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 1.78 \cdot 23^2 \cdot 168 \cdot 1233 / 0.29$$

$$= 5910.502 \text{ cm}^4$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	499281.	79.8750	1.0000	39881.
Face Pressure, Mt	80110.	74.0651	1.0000	5933.
Gasket Load, Mg	78346.	50.2553	1.0000	3937.
Gasket Seating, Matm	809286.	50.2553	1.0000	40672.
Moment Load, Moe	...	...	...	45.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{I}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 33.5 \left[\frac{5911}{(0.384 \cdot 5955 + 5911)} \right] \cdot \left(\frac{79.9}{(1429 - 2 \cdot 79.9)} \right) + 487 \cdot 79.9$$

$$= 44.972 \text{ Kg-m.}$$

Total Moment for Operation, Mo 49797. Kg-m.
 Total Moment for Gasket seating, Mg 40672. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 168.401 mm.
 Hub Ratio, h/h0 = HL / H0 0.545
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.565

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.836
 Factor V per 4.16.5 0.289
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1530 / 1233$$

$$= 1.241$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.24^2 \cdot \log_{10}(1.24))) / (1.24^2 - 1) / (1.24 - 1)$$

$$= 9.121$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.24^2(1 + 8.55246 \cdot \log_{10}(1.24)) - 1) / ((1.0472 + 1.9448 \cdot 1.24^2) \cdot (1.24 - 1))$$

$$= 1.822$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.24^2(1 + 8.55246 \cdot \log_{10}(1.24)) - 1) / (1.36136(1.24^2 - 1) \cdot (1.24 - 1))$$

$$= 10.023$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.24^2 + 1) / (1.24^2 - 1)$$

$$= 4.705$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 10 \cdot 23^2 \cdot 168 / 0.29$$

$$= 3089074.250 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.84 / 168$$

$$= 0.005 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (138 \cdot 0.005 + 1) / 1.82 + 138^3 / 3089074$$

$$= 1.780$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 49797 / 1233) / (1.78 \cdot 36^2)$$

$$= 171.73 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 40672 / 1233) / (1.78 \cdot 36^2)$$

$$= 140.26 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.91 \cdot 49797 / 1233) / (1.78 \cdot 138^2)$$

$$= 22.31 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.91 \cdot 40672 / 1233) / (1.78 \cdot 138^2)$$

$$= 18.22 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (9.12 \cdot 49797 / (138^2 \cdot 1233)) - 4.71 \cdot 22.3$$

$$= 84.18 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (9.12 \cdot 40672 / (138^2 \cdot 1233)) - 4.71 \cdot 18.2$$

$$= 68.75 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (172 + \max(22.3, 84.2)) / 2$$

$$= 127.95 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (140 + \max(18.2, 68.8)) / 2$$

$$= 104.50 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= (Wm_1 + Fa + 4Me/G) / Ab$$

$$= 658325 / 548$$

$$= 117.92 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm_2 / Ab)$$

$$= (263464 / 548)$$

$$= 47.19 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	171.73	207.00	140.26	207.00
Radial Flange	22.31	138.00	18.22	138.00
Tangential Flange	84.18	138.00	68.75	138.00
Maximum Average	127.95	138.00	104.50	138.00
Bolting	117.92	172.00	47.19	172.00

Minimum Required Flange Thickness 130.023 mm.

Estimated M.A.W.P. (Operating) 42.8 bars

Estimated Finished Weight of Flange at given Thk. 780.5 kg.

Estimated Unfinished Weight of Forging at given Thk 1155.7 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 40672 / 1 \cdot 9807 \cdot 0.29 / (1.61 \cdot 201334 \cdot 23^2 \cdot 168 \cdot 0.3)$
 $= 0.692$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 49797 / 1 \cdot 9807 \cdot 0.29 / (1.61 \cdot 194922 \cdot 23^2 \cdot 168 \cdot 0.3)$
 $= 0.875$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 40672 / 1 \cdot 9807 \cdot 0.29 / (1.78 \cdot 201334 \cdot 23^2 \cdot 168 \cdot 0.3)$
 $= 0.627$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 49797 / 1 \cdot 9807 \cdot 0.29 / (1.78 \cdot 194922 \cdot 23^2 \cdot 168 \cdot 0.3)$
 $= 0.793$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.82, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -22 °C

Min Metal Temp. at Required thickness (UCS 66.1) -32 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	169.95	207.00	138.03	207.00
Radial Flange	23.14	138.00	18.79	138.00
Tangential Flange	81.05	138.00	65.83	138.00
Maximum Average	125.50	138.00	101.93	138.00
Bolting	117.92	172.00	47.19	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 127.559 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
BOTTOM HEAD		41.101	28	1	1231	117.3	1
SHELL-1		41.065	24	1	1231	117.3	2
SHELL-2		41.005	24	1	1231	117.3	3
BF-1		41.005	24	1	1231	117.3	4
BF-2		41.005	24	1	1231	117.3	5
TOP HEAD		41.005	28	1	1231	117.3	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.W.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
BOTTOM HEAD		40.9976	43.6	45.6	24	22.6323	1
SHELL-1		40.9976	42.7	44.7	24	23.0472	2
SHELL-2		40.9976	42.8	44.7	24	23.014	3
BF-1		40.9976	42.5	43.3	138.2	134.468	4
BF-2		40.9976	42.8	44.7	138.2	130.023	5
TOP HEAD		40.9976	43.7	45.6	24	22.5812	6

Minimum 41 43.3

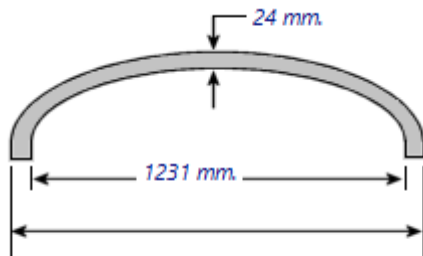
MAWP: 41 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: BOTTOM HEAD

Elliptical Head Design Information:					
Design Pressure	P	41.101 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1231.000 mm.	Actual Stress		129.8 N./mm^2
Spec. Min. Thk t,e, or sa		24.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		19976.4 cm^2	SG of Contents		1.030
Element Volume		303744.1 cm3	Weight of Contents		308.4 Kg
Empty Weight		454.6 Kg	Operating Weight		763.0 Kg
			Comp. All. Ope. Stress		141.7 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.388 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.694 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (41.1 \cdot 1233 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 41.1)$$

$$= 21.6323 + 1.0000 = 22.6323 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.104 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 23) / (1 \cdot 1233 + 0.2 \cdot 23)$$

$$= 43.7 - 0.1 = 43.6 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 24) / (1 \cdot 1231 + 0.2 \cdot 24)$$

$$= 45.6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (41.1 \cdot (1 \cdot 1233 + 0.2 \cdot 23)) / (2 \cdot 0.85 \cdot 23)$$

$$= 129.822 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (41.1 \cdot 616) / (138 \cdot 0.85 - 0.6 \cdot 41.1) + 1$
 $= 23.067 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.073 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 27) / (616 + 0.6 \cdot 27)$
 $= 50.1 - 0.073 = 50 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1233 / (2 \cdot 309))^2) / 6$
 $= 0.997842$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.406 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 24$, $t_r = 21.6$, $c = 1 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -36 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -47 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 28$, $t_r = 22$, $c = 1 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	40.998 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.000 mm.	Basic MDMT	-36 °C
Stress/Thk. Ratio	0.797	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-47 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

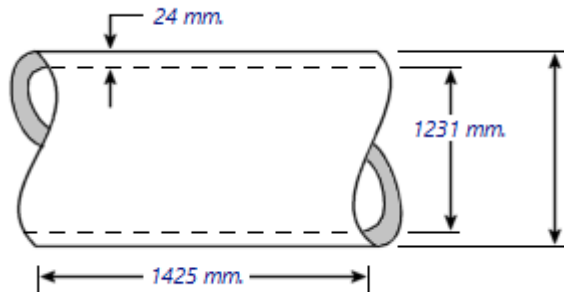
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.388 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.694 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	41.065 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1425.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1231.000 mm.	Actual Stress		132.4 N./mm^2
Spec. Min. Thk t _e , or sa		24.0000 mm.	Spec. Nominal Thk.		24.000 mm.
Surface Area		57257.9 cm^2	SG of Contents		1.030
Element Volume		1696284.5 cm3	Weight of Contents		728.1 Kg
Empty Weight		1045.0 Kg	Operating Weight		1773.1 Kg
			Comp. All. Ope. Stress		141.7 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (41.1 \cdot 616) / (138 \cdot 0.85 - 0.6 \cdot 41.1)$$

$$= 22.0472 + 1.0000 = 23.0472 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.068 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 23) / (616 + 0.6 \cdot 23)$$

$$= 42.8 - 0.068 = 42.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (616 + 0.6 \cdot 24)$$

$$= 44.7 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (41.1 \cdot (616 + 0.6 \cdot 23)) / (0.85 \cdot 23)$$

$$= 132.404 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.912 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24, tr = 22, c = 1 mm., E* = 0.85



Thickness Ratio = $tr * E^* / (tg - c) = 0.81$, Temp. Reduction = 10 °C

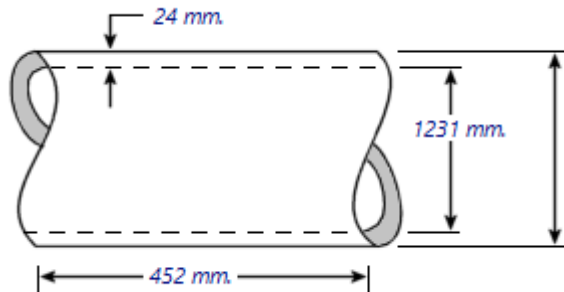
Min Metal Temp. w/o impact per UCS-66, Curve D -36 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	40.998 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.000 mm.	Basic MDMT	-36 °C
Stress/Thk. Ratio	0.813	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-46 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	41.005 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		452.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1231.000 mm.	Actual Stress		132.2 N./mm^2
Spec. Min. Thk t _e , or s _a		24.0000 mm.	Spec. Nominal Thk.		24.000 mm.
Surface Area		18161.8 cm^2	SG of Contents		0.000
Element Volume		538049.6 cm3	Weight of Contents		0.0 Kg
Empty Weight		331.5 Kg	Operating Weight		331.5 Kg
			Comp. All. Ope. Stress		141.7 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (41 \cdot 616) / (138 \cdot 0.85 - 0.6 \cdot 41)$$

$$= 22.0140 + 1.0000 = 23.0140 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 23) / (616 + 0.6 \cdot 23)$$

$$= 42.8 - 0.007 = 42.8 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 24) / (616 + 0.6 \cdot 24)$$

$$= 44.7 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (41 \cdot (616 + 0.6 \cdot 23)) / (0.85 \cdot 23)$$

$$= 132.208 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.912 %

Minimum Design Metal Temperature Results:

Govern. thk, tg = 24, tr = 22, c = 1 mm., E* = 0.85



Thickness Ratio = $tr * E^* / (tg - c) = 0.81$, Temp. Reduction = 10 °C

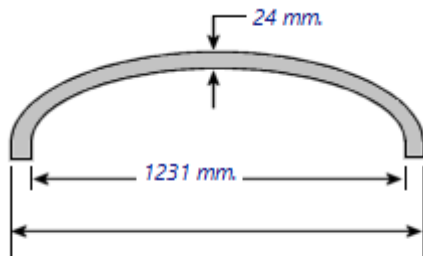
Min Metal Temp. w/o impact per UCS-66, Curve D -36 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	40.998 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.000 mm.	Basic MDMT	-36 °C
Stress/Thk. Ratio	0.813	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-46 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: TOP HEAD

Elliptical Head Design Information:					
Design Pressure	P	41.005 bars	Int. Design Temperature		130 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1231.000 mm.	Actual Stress		129.5 N./mm^2
Spec. Min. Thk t,e, or sa		24.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		19976.4 cm^2	SG of Contents		0.000
Element Volume		303744.1 cm3	Weight of Contents		0.0 Kgf
Empty Weight		454.6 Kgf	Operating Weight		454.6 Kgf
			Comp. All. Ope. Stress		141.7 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.388 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.694 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (41 \cdot 1233 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 41)$$

$$= 21.5812 + 1.0000 = 22.5812 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 23) / (1 \cdot 1233 + 0.2 \cdot 23)$$

$$= 43.7 - 0.007 = 43.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 24) / (1 \cdot 1231 + 0.2 \cdot 24)$$

$$= 45.6 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (41 \cdot (1 \cdot 1233 + 0.2 \cdot 23)) / (2 \cdot 0.85 \cdot 23)$$

$$= 129.517 \text{ N./mm}^2$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (41 \cdot 616) / (138 \cdot 0.85 - 0.6 \cdot 41) + 1$
 $= 23.014 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.007 bars
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 0.85 \cdot 27) / (616 + 0.6 \cdot 27)$
 $= 50.1 - 0.007 = 50 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (1233 / (2 \cdot 309))^2) / 6$
 $= 0.997842$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.406 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 24$, $t_r = 21.6$, $c = 1 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -36 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -47 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 28$, $t_r = 22$, $c = 1 \text{ mm.}$, $E^* = 0.85$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	40.998 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.000 mm.	Basic MDMT	-36 °C
Stress/Thk. Ratio	0.797	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-47 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 15.388 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 7.694 mm.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	53.297 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	53.297 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	56.107 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	45.097 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	58.627 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	58.627 bars
App 27-4	= M.A.W.P.	40.998 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	170.3	234.0	0.728	56.23
SHELL-1	173.6	234.0	0.742	56.23
SHELL-2	173.6	234.0	0.742	56.23
TOP HEAD	170.3	234.0	0.728	56.23

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-2"	Nozzle	118.00	118.00	1.000
N1-4"	Nozzle	138.00	138.00	1.000
K1-2"	Nozzle	115.00	115.00	1.000
K2-2"	Nozzle	115.00	115.00	1.000
K3-2"	Nozzle	115.00	115.00	1.000
K4-2"	Nozzle	115.00	115.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
N2-4"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
BF-1	138.00	138.00	1.000
BF-2	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-2"	11.89	216.00	0.055
N1-4"	26.64	225.00	0.118
K1-2"	11.89	153.00	0.078
K2-2"	11.89	153.00	0.078
K3-2"	11.89	153.00	0.078
K4-2"	11.89	153.00	0.078
M1-24"	74.79	225.00	0.332
N2-4"	26.64	225.00	0.118

Elements Suitable for Test Pressure.



Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	53.297 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	53.297 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	56.148 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	45.097 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	58.627 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	58.627 bars
App 27-4	= M.A.W.P.	40.998 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 41 • 1
 = 53.297 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
BOTTOM HEAD	169.3	234.0	0.723	53.60
SHELL-1	172.7	234.0	0.738	53.56
SHELL-2	172.2	234.0	0.736	53.42
TOP HEAD	168.5	234.0	0.720	53.33

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-2"	Nozzle	118.00	118.00	1.000
N1-4"	Nozzle	138.00	138.00	1.000
K1-2"	Nozzle	115.00	115.00	1.000
K2-2"	Nozzle	115.00	115.00	1.000
K3-2"	Nozzle	115.00	115.00	1.000
K4-2"	Nozzle	115.00	115.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
N2-4"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
BOTTOM HEAD	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000
BF-1	138.00	138.00	1.000
BF-2	138.00	138.00	1.000
TOP HEAD	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-2"	11.34	216.00	0.052
N1-4"	25.37	225.00	0.113
K1-2"	11.33	153.00	0.074
K2-2"	11.33	153.00	0.074
K3-2"	11.33	153.00	0.074
K4-2"	11.33	153.00	0.074



M1-24"	71.24	225.00	0.317
N2-4"	25.26	225.00	0.112

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	1279	23	0.0024976	108.464
20	30	2029.58	1279	23	0.0019992	103.748
30	40	2029.58	1279	23	0.0019992	103.748
40	50	No Calc	...	137.2	No Calc	No Calc
50	60	No Calc	...	137.2	No Calc	No Calc
60	70	No Calc	1279	23	0.0024976	108.464

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	24	No Calc	...	21.7
20	30	24	No Calc	...	24.9
30	40	24	No Calc	...	24.9
40	50	138.2	95.885	...	No Calc
50	60	138.2	77.6732	...	No Calc
60	70	24	No Calc	...	21.7

Minimum

21.7

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2029.58	No Calc	No Calc	No Calc
30	40	2029.58	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	454.593	303744	438.358	305130	31.8215
20	30	1045.03	1696284	1002.28	1701800	73.1518
30	40	331.475	538050	317.917	539799	23.2033
40	50	780.451	274321	773.552	274677	54.6316
50	60	780.451	274321	773.552	274677	54.6316
60	70	454.593	303744	438.358	305130	31.8215
Total		3846	3390464.75	3744	3401213.25	269

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	308.411	...	-153.875	...	Liquid: 10
10	Insl	74.6141	...	-128.875	...	Insulation
10	Lini	48.0016	...	-128.875	...	SS 316L CLAD
10	Nozl	37.9067	...	-885.341	...	N3-2"
10	Wght	...	...	-307	...	Anti Swirl Baff
20	Liqd	728.052	...	300	...	Liquid: 20
20	Insl	171.524	...	712.5	...	Insulation
20	Lini	132.095	...	712.5	...	Lining
20	Nozl	35.0522	755.505	1200	...	N1-4"
20	Nozl	13.7361	735.502	850	...	K1-2"
20	Nozl	13.7361	735.502	450	...	K2-2"
20	Nozl	13.7361	-735.502	600	...	K3-2"
20	Nozl	13.7361	-735.502	250	...	K4-2"
20	Nozl	968.145	...	850	830.498	M1-24"
20	Legs	367.817	...	-665	...	LEGS
20	Wght	50	...	850	1275	Davit Weight-M1
20	Wght	814	...	710	...	Clip Weight
30	Insl	54.4062	...	226	...	Insulation-3
30	Lini	41.8998	...	226	...	Lining
30	Wght	26	...	276	...	Demisting Pad
30	Wght	32	...	276	...	Demisting Suppo
40	Insl	25.4844	...	115	...	Insulation
40	Lini	21.3207	...	115	...	Lining
50	Insl	25.4844	...	115	...	Insulation
50	Lini	21.3207	...	115	...	Lining
60	Insl	74.6141	...	178.875	...	Insulation
60	Lini	48.0016	...	178.875	...	Lining
60	Nozl	35.7071	...	485.755	...	N2-4"
30	Trun	64.2636	...	226	...	Trunnions

Total Weight of Each Detail Type:

Liquid	1036.5
Insulation	426.1
Cladding & Overlay	312.6
Nozzles	1131.8
Legs	367.8
Weights	922.0
Trunnions	64.3

Sum of the Detail Weights 4261.1 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	4115.9	4115.9	4115.9	4115.9	4115.9	4115.9



Cladding	312.6	312.6	312.6	312.6	312.6	312.6
Nozzles	1131.8	1131.8	1131.8	1131.8	1131.8	1131.8
Legs	367.8	367.8	367.8	367.8	367.8	367.8
Wld Weights	896.0	896.0	896.0	896.0	896.0	896.0
Trunnions	64.3	64.3	64.3	64.3	64.3	64.3
Insulation	...	...	...	426.1	426.1	426.1
Empty Weights	...	...	...	26.0	26.0	...
Ope Weights	...	...	...	...	...	26.0
Ope. Liquid	...	...	...	...	...	1036.5
Test Liquid	...	3388.4	...	...	...	...
Totals	6888.3	10276.7	6888.3	7340.5	7340.5	8376.9

Field Installation Options:

Insulation installed before lifting.

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	6888.3 kg.
Shop Test Wt.	- Fabricated weight + water (full)	10276.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	6888.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	7340.5 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	7340.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	8376.9 kg.
Field Test Wt.	- Empty weight + test medium (full)	10663.8 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	2723.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	3713.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	2765.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	19976.4	18359.9
20	30	57257.9	55109
30	40	18161.8	17480.2
40	50	16890.1	8894.79
50	60	16890.1	8894.79
60	70	19976.4	18359.9

Total	149152.8	127098.7 cm2
Total	160.5	136.8 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	524.322	955.348	950.496	...	646.937
20	Legs	533.389	714.384	884.068	804.058	586.656
Legs	30	2506.93	3357.61	4155.12	...	2757.28
30	40	476.942	573.248	1084.97	...	573.248
40	50	835.083	881.888	1156.04	...	881.888
50	60	835.083	881.888	1156.04	...	881.888
60	70	522.122	644.738	948.296	...	644.738



Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	Legs	-646.937	-955.348	-950.496
Legs	30	5739.04	6339.37	8500.47
30	40	2981.76	2981.76	4345.35
40	50	2408.51	2408.51	3260.38
50	60	1526.63	1526.63	2104.34
60	70	644.738	644.738	948.296

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	Legs	804.058	804.058	804.058
Legs	30	...	...	...
30	40	...	...	...
40	50	...	...	...
50	60	...	...	...
60	70	...	...	...

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3-2"	91.42	102.00	130	600	GR 1.1	76.295	60.836	91.421	60.836
N1-4"	91.42	102.00	130	600	GR 1.1	58.525	78.606	91.421	78.606
K1-2"	65.70	82.30	130	600	GR 2.3	...	...	...	...
K2-2"	65.70	82.30	130	600	GR 2.3	...	...	...	...
K3-2"	65.70	82.30	130	600	GR 2.3	...	...	...	...
K4-2"	65.70	82.30	130	600	GR 2.3	...	...	...	...
M1-24"	45.74	51.00	130	300	GR 1.1	...	...	...	...
N2-4"	91.42	102.00	130	600	GR 1.1	58.525	78.606	91.421	78.606
Min Rating	45.740	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N3-2"	77.851	0.50	72	0	40.998	60.836	Pass
N1-4"	135.001	0.50	288	0	40.998	78.606	Pass
N2-4"	135.001	0.50	288	0	40.998	78.606	Pass

Evaluating Flange on Nozzle [N3-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(72) + 4 \cdot \text{abs}(0)77.851 \leq \pi \cdot 77.851^3 \cdot ((91.421 - 40.998) + 500 \cdot 91.421)$$

$$= 0.11533E+04 \leq 0.14532E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(91.421, 0.500 \cdot 91.421 + 91.421 - 1153.3 / (\pi \cdot 77.851^3))$$

$$= \min(91.421, 60.836)$$

$$= 60.836 \text{ bars}$$

Evaluating Flange on Nozzle [N1-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(288) + 4 \cdot \text{abs}(0)135.001 \leq \pi \cdot 135.001^3 \cdot ((91.421 - 40.998) + 500 \cdot 91.421)$$

$$= 0.46132E+04 \leq 0.75778E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(91.421, 0.500 \cdot 91.421 + 91.421 - 4613.2 / (\pi \cdot 135.001^3))$$

$$= \min(91.421, 78.606)$$

$$= 78.606 \text{ bars}$$

Evaluating Flange on Nozzle [N2-4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(288) + 4 \cdot \text{abs}(0)135.001 \leq \pi \cdot 135.001^3 \cdot ((91.421 - 40.998) + 500 \cdot 91.421)$$

$$= 0.46132E+04 \leq 0.75778E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(91.421, 0.500 \cdot 91.421 + 91.421 - 4613.2 / (\pi \cdot 135.001^3))$$

$$= \min(91.421, 78.606)$$



= 78.606 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 17.2896 Hz.

The Natural Frequency for the Vessel (Ope...) is 16.6814 Hz.

The Natural Frequency for the Vessel (Filled) is 14.3119 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.97) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	16.681 Hz
Natural Frequency of Vessel (Empty)	17.290 Hz
Natural Frequency of Vessel (Test)	14.312 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.518
Structure Height to Diameter ratio	2.061
Height to top of Structure	2853.250 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 264 + 0 = 264 \text{ mm.}$
 $= 0.87 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (0.87 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$$

$$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16 \cdot 0.85 \cdot 0.52 \cdot 7.69$$

$$= 54.2 \text{ lbs. } [24.6] \text{ Kgf}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
BOTTOM HEAD	263.7	0.000	0.000	0.000	0.849	1.000	78.120
SHELL-1	1070.2	0.000	0.000	0.000	0.849	1.000	78.120
SHELL-2	2008.8	0.000	0.000	0.000	0.849	1.000	78.120
BF-1	2354.2	0.000	0.000	0.000	0.849	1.000	78.120
BF-2	2584.3	0.000	0.000	0.000	0.849	1.000	78.120
TOP HEAD	2897.3	0.000	0.000	0.000	0.849	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm.
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm.
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm.
 Lc - Total length of conical section(s) of vessel, mm.
 tb - Uncorroded plate thickness at bottom of vessel, mm.
 V30 - Design Wind Speed provided by user, Km/hr
 Vc - Critical wind velocity, Km/hr
 Vw - Maximum wind speed at top of structure, Km/hr
 W - Total corroded weight of structure, Kgf
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
 Z - Maximum amplitude of vibration at top of vessel, mm.
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.32037\text{E-}06$ (High); $0.32037\text{E-}06 < 0.40047\text{E-}06$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0/2442 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[4.68/8.01^2] \cdot 10^4 = 730$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.40047\text{E-}06$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 8267 / (2442 \cdot 1472^2)$$

$$= 0.15626\text{E-}05$$

Since Vp is $> 0.40047\text{E-}06$ no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping



ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
Anti Swirl Ba	0.00	-307.00	78.12	0.00
Davit Weight-	0.00	900.00	78.12	0.00
Clip Weight	0.00	760.00	78.12	0.00
Demisting Pad	0.00	1751.00	78.12	0.00
Demisting Sup	0.00	1751.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 16.6814 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	263.746	1913.87	7146.16	78.12	27.0216
20	30	1070.25	1913.87	27272.7	78.12	103.125
30	40	2008.75	1913.87	8650.69	78.12	32.7107
40	50	2354.25	1850.03	4255.07	78.12	16.0896
50	60	2584.25	1850.03	4255.07	78.12	16.0896
60	70	2897.25	1913.87	7146.16	78.12	27.0216

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.3 * 0.15 = 0.19 \\
 SM1 &= Fv * S1 = 1.5 * 0.065 = 0.097 \\
 SDS &= 2/3 * Sms = 2/3 * 0.19 = 0.13 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.097 = 0.065
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(140.75)$
 $= 0.145$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/16.7)$
 $= 0.060$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.14, 0.06) \text{ per 12.8.2}$
 $= 0.060$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.13 / (1.25 / 1.25)$
 $= 0.127$

Maximum value of C_s per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / I_e))$
 $= 0.065 / (0.06 (1.25 / 1.25))$
 $= 1.084$

Minimum value of C_s per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot I_e, 0.030)$
 $= \max(0.044 \cdot 0.13 \cdot 1.25, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= C_s \cdot W$ (Equation 12.8-1):
 $= 0.13 \cdot 8009$
 $= 1020.359$ Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_{xk} / (\text{Sum of } W_i \cdot h_{ik})$ and k is an exponent which is related



to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 16.6814 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	25	955.348	2.54821	0.48309	...
20	Legs	300	714.384	22.8658	5.89732	...
Legs	30	887.5	3357.61	317.93	81.9974	...
30	40	1701	573.248	104.035	29.8992	...
40	50	2046.5	881.888	192.556	62.984	...
50	60	2276.5	881.888	214.197	70.0626	...
60	70	2416.5	644.738	166.227	46.4995	...



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	448.003	...	...	...	...
20	Legs	125	27.0216	2.54821	5.35047	0.50456
Legs	30	587.5	222.058	1020.36	203.189	1468.8
30	40	1401	91.9114	677.016	59.6005	490.508
40	50	1746.5	59.2008	572.981	25.182	205.424
50	60	1976.5	43.1112	380.425	13.4159	95.7805
60	70	2289.5	27.0216	166.227	5.35047	32.9142

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	...	0.60887	0.60887	0.00049893
20	Legs	27.0216	0.60887	0.60889	0.00049909
Legs	30	222.058	0.60908	0.6094	0.00049954
30	40	91.9114	0.60954	0.60969	0.0004996
40	50	59.2008	0.60976	0.60984	0.0004996
50	60	43.1112	0.60992	0.60999	0.0004996
60	70	27.0216	0.61001	0.61003	0.0004996

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	573.121	3582.01
20	30	573.121	3582.01
30	40	573.121	3582.01
40	50	554.004	3462.52
50	60	554.004	3462.52
60	70	573.121	3582.01

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 12.208 Actual deflection : 0.610 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	946.248	907.543	29142056	27993792
20	30	946.248	907.543	29142056	27993792
30	40	946.248	907.543	29142056	27993792
40	50	946.248	907.543	29142056	27993792
50	60	946.248	907.543	29142056	27993792
60	70	946.248	907.543	29142056	27993792

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	140.76	238.68	-141.708	-141.708
20	Legs	140.76	238.68	-141.708	-141.708
Legs	30	140.76	238.68	-141.708	-141.708
30	40	140.76	238.68	-141.708	-141.708
40	50	140.76	229.5	-141.708	-141.708
50	60	140.76	229.5	-141.708	-141.708
60	70	140.76	238.68	-141.708	-141.708

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	54.1381	...	70.3676
20	30	54.1381	...	70.3676
30	40	54.1381	...	70.3676
40	50	...	...	...
50	60	...	...	...
60	70	54.1381	...	70.3676

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.069907	0.10323	0.10271	0.28167	0.28167
Legs	30	-0.62016	-0.60635	-0.62016	...	...
30	40	-0.32221	-0.32221	-0.32221	...	...
40	50	-0.26026	-0.26026	-0.26026	...	...
50	60	-0.16497	-0.16497	-0.16497	...	...
60	70	-0.06967	-0.06967	-0.06967	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.28167	0.0018743	0.00017676	0.00093717	...
Legs	30	...	0.07118	0.51454	0.03559	...
30	40	...	0.020879	0.17183	0.010439	...
40	50	...	0.0088216	0.071963	0.0044108	...
50	60	...	0.0046998	0.033553	0.0023499	...
60	70	...	0.0018743	0.01153	0.00093717	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	0.33509E-04
Legs	30	...	...	...	0.15556
30	40	...	...	...	0.052805
40	50	...	...	...	0.021724
50	60	...	...	...	0.0097947
60	70	...	...	...	0.0032254

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	...
20	Legs	...	...
Legs	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	Legs	...
Legs	30	...
30	40	...
40	50	...
50	60	...
60	70	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.71	0.0000	-0.0000
20	0.35	140.76	-0.21	141.71	0.0025	0.0015
20		140.76	-0.69	141.71		0.0049
30		140.76	-0.34	141.71		0.0024
60		140.76	-0.07	141.71		0.0005

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.71	0.0000	-0.0000
20	0.35	140.76	-0.21	141.71	0.0025	0.0015
20		140.76	-0.78	141.71		0.0055
30		140.76	-0.38	141.71		0.0026
60		140.76	-0.07	141.71		0.0005

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.71	0.0000	-0.0000
20	0.39	140.76	-0.18	141.71	0.0027	0.0013
20		140.76	-0.68	141.71		0.0048
30		140.76	-0.34	141.71		0.0024



60 140.76 -0.07 141.71 0.0005

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.71	0.0000	-0.0000
20	0.39	140.76	-0.18	141.71	0.0027	0.0013
20		140.76	-1.12	141.71		0.0079
30		140.76	-0.49	141.71		0.0035
60		140.76	-0.08	141.71		0.0006

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	141.71	0.0000	-0.0000
20	0.39	238.68	-0.18	141.71	0.0016	0.0013
20		238.68	-0.66	141.71		0.0046
30		238.68	-0.33	141.71		0.0023
60		238.68	-0.07	141.71		0.0005

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	141.71	0.0000	-0.0000
20	0.38	238.68	-0.18	141.71	0.0016	0.0013
20		238.68	-0.62	141.71		0.0044
30		238.68	-0.32	141.71		0.0023
60		238.68	-0.07	141.71		0.0005

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	54.14	140.76		141.71	0.3846	
20	54.52	140.76		141.71	0.3874	
20	53.60	140.76		141.71	0.3808	
30	53.84	140.76		141.71	0.3825	
60	54.07	140.76		141.71	0.3841	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	54.14	140.76		141.71	0.3846	
20	54.52	140.76		141.71	0.3873	
20	54.05	140.76		141.71	0.3840	
30	53.99	140.76		141.71	0.3835	
60	54.08	140.76		141.71	0.3842	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.71	0.0000	-0.0000
20	0.39	140.76	-0.18	141.71	0.0027	0.0013
20		140.76	-0.68	141.71		0.0048
30		140.76	-0.34	141.71		0.0024
60		140.76	-0.07	141.71		0.0005

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.71	0.0000	-0.0000
20	0.39	140.76	-0.18	141.71	0.0027	0.0013
20		140.76	-1.12	141.71		0.0079
30		140.76	-0.49	141.71		0.0035



60	140.76	-0.08	141.71	0.0006
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Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	70.37	238.68		141.71	0.2948	
20	70.75	238.68		141.71	0.2964	
20	69.78	238.68		141.71	0.2924	
30	70.06	238.68		141.71	0.2935	
60	70.30	238.68		141.71	0.2945	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	70.37	238.68		141.71	0.2948	
20	70.75	238.68		141.71	0.2964	
20	69.75	238.68		141.71	0.2922	
30	70.05	238.68		141.71	0.2935	
60	70.30	238.68		141.71	0.2945	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	54.14	140.76		141.71	0.3846	
20	54.49	140.76		141.71	0.3871	
20	53.52	140.76		141.71	0.3802	
30	53.82	140.76		141.71	0.3823	
60	54.07	140.76		141.71	0.3841	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	54.14	140.76		141.71	0.3846	
20	54.24	140.76		141.71	0.3853	
20	53.56	140.76		141.71	0.3805	
30	53.83	140.76		141.71	0.3824	
60	54.07	140.76		141.71	0.3841	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	54.14	140.76		141.71	0.3846	
20	54.52	140.76		141.71	0.3873	
20	53.53	140.76		141.71	0.3803	
30	53.82	140.76		141.71	0.3823	
60	54.07	140.76		141.71	0.3841	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	54.14	140.76		141.71	0.3846	
20	54.49	140.76		141.71	0.3871	
20	53.52	140.76		141.71	0.3802	
30	53.82	140.76		141.71	0.3823	
60	54.07	140.76		141.71	0.3841	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	141.71	0.0000	-0.0000
20	0.38	140.76	-0.18	141.71	0.0027	0.0013
20		140.76	-0.61	141.71		0.0043
30		140.76	-0.32	141.71		0.0023



60 140.76 -0.07 141.71 0.0005

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	54.14	117.30		118.09	0.4615	
20	54.52	117.30		118.09	0.4648	
20	53.53	117.30		118.09	0.4564	
30	53.82	117.30		118.09	0.4588	
60	54.07	117.30		118.09	0.4609	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	117.30	0.00	118.09	0.0000	-0.0000
20	0.38	117.30	-0.18	118.09	0.0033	0.0015
20		117.30	-0.61	118.09		0.0051
30		117.30	-0.32	118.09		0.0027
60		117.30	-0.07	118.09		0.0006

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	54.14	140.76		141.71	0.3846	
20	54.52	140.76		141.71	0.3873	
20	53.53	140.76		141.71	0.3803	
30	53.82	140.76		141.71	0.3823	
60	54.07	140.76		141.71	0.3841	

Absolute Maximum of the all of the Stress Ratio's 0.4648

Governing Element: SHELL-1

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
BOTTOM HEAD	54.1	...	...	...	54.1	+ Tens
BOTTOM HEAD	54.1	...	...	...	...	- Comp
SHELL-1	54.1	0.1	0.3	...	54.5	+ Tens
SHELL-1	54.1	0.1	-0.3	...	...	- Comp
SHELL-1	54.1	-0.6	...	...	53.5	+ Tens
SHELL-1	54.1	-0.6	...	...	...	- Comp
SHELL-2	54.1	-0.3	...	...	53.8	+ Tens
SHELL-2	54.1	-0.3	...	...	...	- Comp
TOP HEAD	54.1	-0.1	...	...	54.1	+ Tens
TOP HEAD	54.1	-0.1	...	...	...	- Comp

Shop/Field Installation Options :

Insulation is installed in the Field before being lifted.
 Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	200.066 mm.
Center of Gravity of the Insulation	1210.135 mm.
Center of Gravity of the Lining	1219.805 mm.
Center of Gravity of the Nozzles	898.254 mm.
Center of Gravity of the Legs	-615.000 mm.
Center of Gravity of the Added Weights (Operating)	829.933 mm.
Center of Gravity of the Added Weights (Empty)	829.933 mm.
Center of Gravity of the Trunnions	1701.000 mm.
Center of Gravity of Bare Shell New and Cold	1519.385 mm.
Center of Gravity of Bare Shell Corroded	1527.592 mm.
Vessel CG in the Operating Condition	1075.216 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	1199.462 mm.
Vessel CG in the Test Condition	1206.403 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated empty weight

Total Effective Length of Vessel for this analysis	2437.00 mm.
Total vessel weight (No Liquid) Twt	7340.45 Kg
Impact weight multiplication factor Imp	2.00
Design lifting weight, DWT = Imp * Twt	14680.91 Kg
Elevation of the Tailing Lug (bottom)	0.00 mm.
Elevation of the Lifting Lug (top)	1701.00 mm.
Design Reaction force at the tailing lug	4328.65 Kg
Design Reaction force at the Trunnions	10352.26 Kg
CG distance during field lifting operation	1199.46 mm.
CG distance from tailing lug	1199.46 mm.
CG distance from the nearer lifting lug/trunnion	501.54 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	0.87	1655.80	118.63 (UG-23)
Shear	0.60	1746.20	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kg	Shear Stress N./mm ²
0.00	0.0	0.0	888.8	0.1
50.00	147.6	0.0	1015.9	0.1
1475.00	1956.5	0.7	3142.5	0.3
1927.00	1536.1	0.5	3383.3	0.4
2157.00	704.6	0.2	2227.0	0.2
2387.00	238.9	0.1	202.5	0.0

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0074 at elevation 1656 mm. - Must be <=1



Note: The rigging analysis is performed using a uniformly distributed load.

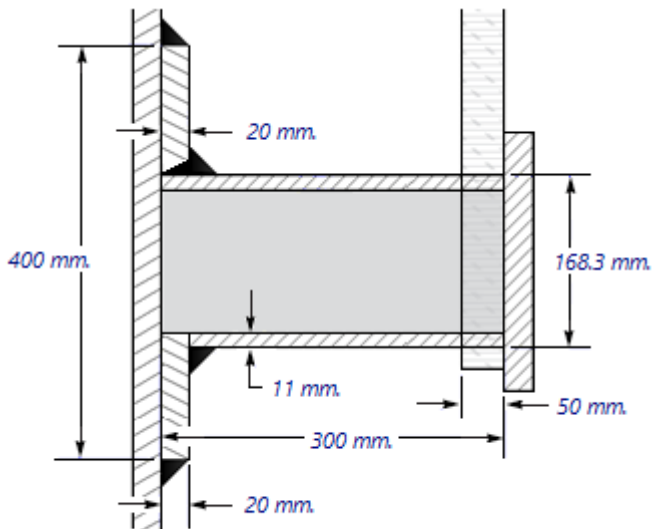
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Trunnion Analysis:

Input Data:

Lifting Lug Material	SA-516 70
Lifting Lug Yield Stress	Yield 262.00 N./mm ²
Trunnion Outside Diameter	Tod 168.300 mm.
Trunnion Thickness	Tthk 10.9700 mm.
Trunnion Length	Tl 300.0000 mm.
Reinforcing Pad Used	Yes
Pad Outside Diameter	Dp 400.000 mm.
Pad Thickness	Thkp 20.0000 mm.
Pad Edge Fillet Weld Thk	Tfp 20.0000 mm.
Bail/Sling Width	Tt 50.0000 mm.
Sling Angle from Horizontal	0.0 degrees
Occasional Load Factor	Occfac 1.0
Impact Factor	Impfac 2.00
Number of Lugs in Group	2



Trunnion may be inserted or abutting

Trunnion Force and Stress Results for different Lift Angles (Kgf&N./mm²):

Bending Stress Allowable:	172.9 N./mm ²
Shear Stress Allowable:	104.8 N./mm ²
Bearing Stress Allowable:	196.5 N./mm ²

Angle	Trunnion Lifting Load	Bending Stress	Shear Stress	Bearing Stress	Unity Check	Pass Fail
0	5176.1	69.7	9.4	6.0	0.492	Pass
5	5253.1	70.7	9.5	6.1	0.500	Pass
10	5325.9	71.7	9.6	6.2	0.507	Pass
15	5395.8	72.6	9.8	6.3	0.513	Pass
20	5464.0	73.6	9.9	6.4	0.520	Pass
25	5531.7	74.5	10.0	6.4	0.526	Pass
30	5599.8	75.4	10.1	6.5	0.533	Pass
35	5669.4	76.3	10.3	6.6	0.539	Pass
40	5741.7	77.3	10.4	6.7	0.546	Pass
45	5818.0	78.3	10.5	6.8	0.553	Pass
50	5899.9	79.4	10.7	6.9	0.561	Pass
55	5989.5	80.6	10.8	7.0	0.570	Pass
60	6089.5	82.0	11.0	7.1	0.579	Pass
65	6203.8	83.5	11.2	7.2	0.590	Pass
70	6337.6	85.3	11.5	7.4	0.603	Pass
75	6499.4	87.5	11.8	7.6	0.618	Pass
80	6702.2	90.2	12.1	7.8	0.637	Pass



85	6968.5	93.8	12.6	8.1	0.663	Pass
90	7340.5	98.8	13.3	8.6	0.698	Pass

Trunnion Calculation Results:

Trunnion Lifting Load [Tlf]:
 = Resultant Load • Impact Factor
 = 3670 • 2
 = 7340.5 Kgf

Trunnion Section Modulus [Z]:
 = $\pi (T_{od}^4 - T_{id}^4) / (32 \cdot T_{od})$
 = $\pi (168^4 - 146^4) / (32 \cdot 168)$
 = 200333.4 mm.³

Trunnion Bending Moment [Tmom]:
 = $(T_l - 0.5 \cdot T_t) / 9.807 \cdot T_{lf}$
 = $(300 - 0.5 \cdot 50) / 9.807 \cdot 7340$
 = 2018.7 Kg-m.

Trunnion Calculated Bending Stress [Sbc]:
 = Tmom • 9.807/Z
 = 2019 • 9.807/200333
 = 98.82 N./mm [Passed] Sbc < min(0.66 • SY • Occfac, 0.75 • Sy) (172.92)

Trunnion Cross Section Area [Txsa]:
 = $\pi/4 (T_{od}^2 - T_{id}^2)$
 = $\pi/4 (168^2 - 146^2)$
 = 54.2 cm²

Trunnion Calculated Shear Stress [Ssc]:
 = Tlf/Txsa
 = 7340/54.2
 = 13.28 N./mm [Passed] Ssc < 0.4 • Sy • Occfac (104.80)

Trunnion Unity Check [Uchk]:
 = Sbc/Sba + Ssc/Ssa
 = 98.8/173 + 13.3/105
 = 0.698 [Passed]

Weld Section Modulus at Edge of Trunnion [Sw]:
 = $\pi (T_{od}/2)^2$
 = $\pi (168/2)^2$
 = 22246.3 mm.²

Shear Stress in Weld due to Bending Moment [fs]:
 = Tmom/Sw
 = 2019/22246
 = 90.740 Kgf/mm.

Required Fillet Weld Leg Size at Edge of Trunnion [Wsr]:
 = max(10mm, fs/(0.4 • Sy))
 = max(10mm, 90.7/(0.4 • 262))
 = 10.000 mm.

Trunnion Bearing Contact Area [Tbca]:
 = T_{od} • T_t
 = 168 • 50
 = 84.2 cm²

Trunnion Bearing Stress [Sbec]:
 = Tlf/Tbca
 = 7340/84.2
 = 8.55 N./mm [Passed] Sbec < min(0.75 • SY • Occfac, 0.9 • Sy) (196.50)

Summary of Results:

Trunnion Unity Check Value: 0.6982 Must be less than or equal to 1

Comparison of Stress Values versus Allowables: (N./mm²)

Type of Stress	Actual	Allowable	Pass/Fail
Bending Stress	98.82	172.92	Pass
Shear Stress	13.28	104.80	Pass
Bearing Stress	8.55	196.50	Pass

Results of local shell stress analysis due to loads on the trunnion:

This following Stress Analysis assumes that the Trunnion passes through the pad and is welded to the shell via a groove weld. Additionally the re-pad is welded to the shell at the outside edge via a fillet weld.

In the local stress results below, the 'Nozzle' is the 'Trunnion'.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

Loads on shell and WRC 107 results at different Lift Angles (Kgf & Kg-m.):

Angle	PT	PL	MC	ML	Rat1	Rat2	P/F
0	5176.1	0.0	1423.5	0.0	0.149	0.149	Pass
5	5233.1	457.8	1439.1	125.9	0.151	0.151	Pass
10	5245.0	924.8	1442.4	254.3	0.151	0.151	Pass
15	5211.9	1396.5	1433.3	384.1	0.150	0.150	Pass
20	5134.5	1868.8	1412.0	513.9	0.148	0.148	Pass
25	5013.4	2337.8	1378.7	642.9	0.145	0.145	Pass
30	4849.6	2799.9	1333.7	770.0	0.140	0.140	Pass
35	4644.1	3251.8	1277.2	894.3	0.135	0.134	Pass
40	4398.4	3690.7	1209.6	1015.0	0.128	0.127	Pass
45	4113.9	4113.9	1131.4	1131.4	0.120	0.119	Pass
50	3792.4	4519.6	1042.9	1242.9	0.114	0.110	Pass
55	3435.4	4906.3	944.8	1349.3	0.123	0.100	Pass
60	3044.8	5273.7	837.3	1450.3	0.131	0.089	Pass
65	2621.8	5622.5	721.0	1546.2	0.140	0.077	Pass
70	2167.6	5955.4	596.1	1637.8	0.148	0.079	Pass
75	1682.2	6277.9	462.6	1726.5	0.155	0.083	Pass
80	1163.8	6600.4	320.1	1815.1	0.163	0.087	Pass
85	607.3	6942.0	167.0	1909.1	0.172	0.091	Pass
90	0.0	7340.5	0.0	2018.7	0.181	0.096	Pass

- PT - Transverse load per lug
- PL - Longitudinal load per lug
- MC - Circumferential moment in shell
- ML - Longitudinal moment in shell
- Rat1 - Computed WRC 107 stress ratio at trunnion edge
- Rat2 - Computed WRC 107 stress ratio at pad edge

Worst case combination loads were at: 90.0 degrees from horizontal.

Input Echo, WRC107/537 Item 1, Description: Trunnion

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	0.0000	mm.
Vessel Diameter	Dv	1231.000	mm.
Vessel Thickness	Tv	24.000	mm.
Design Temperature	T1	37.8	C
Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	168.300	mm.
Nozzle Thickness	Tn	10.970	mm.
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²

Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Thickness of Reinforcing Pad	Tpad	20.000	mm.
Diameter of Reinforcing Pad	Dpad	440.000	mm.
Design Internal Pressure	Dp	0.000	bars
Include Pressure Thrust	No		

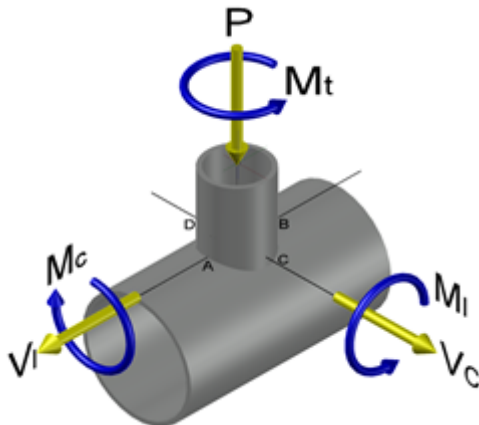
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0	Kgf
Longitudinal Shear (SUS)	VL	7340.5	Kgf
Circumferential Shear (SUS)	Vc	0.0	Kgf
Circumferential Moment (SUS)	Mc	0.0	Kg-m.
Longitudinal Moment (SUS)	ML	2018.7	Kg-m.
Torsional Moment (SUS)	Mt	0.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Worst case combination loads were at: 90.0 degrees from horizontal.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.3 + 2 \cdot 1.65 \cdot \sqrt{627.5(24.0 - 0)}$$

$$= 573.273 \text{ mm.}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	0.0	Kgf
Longitudinal Shear	VL	7340.5	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	2018.7	Kg-m.
Torsional Moment	MT	0.0	Kg-m.

Dimensionless Parameters used : Gamma = 14.49

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
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N(PHI) / (P/Rm)	0.116	4C	2.719	(A,B)
N(PHI) / (P/Rm)	0.116	3C	2.590	(C,D)
M(PHI) / (P)	0.116	2C1	0.117	(A,B)
M(PHI) / (P)	0.116	1C	0.151	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.116	3A	0.314	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.116	1A	0.102	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.116	3B	1.163	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.116	1B	0.054	(A,B,C,D)
N(x) / (P/Rm)	0.116	3C	2.590	(A,B)
N(x) / (P/Rm)	0.116	4C	2.719	(C,D)
M(x) / (P)	0.116	1C1	0.154	(A,B)
M(x) / (P)	0.116	2C	0.117	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.116	4A	0.430	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.116	2A	0.058	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.116	4B	0.296	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.116	2B	0.087	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-11.1	-11.1	11.1	11.1	0.0	0.0	0.0	0.0
Circ. Bend. ML		-44.7	44.7	44.7	-44.7	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-55.8	33.5	55.8	-33.5	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML		-2.8	-2.8	2.8	2.8	0.0	0.0	0.0	0.0
Long. Bend. ML		-72.2	72.2	72.2	-72.2	0.0	0.0	0.0	0.0
Tot. Long. Str.		-75.1	69.4	75.1	-69.4	0.0	0.0	0.0	0.0
Shear VC		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-6.2	-6.2	6.2	6.2
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		0.0	0.0	0.0	0.0	-6.2	-6.2	6.2	6.2
Str. Int.		75.1	69.4	75.1	69.4	12.4	12.4	12.4	12.4

Dimensionless Parameters used : Gamma = 26.15

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.307	4C	3.197	(A,B)
N(PHI) / (P/Rm)	0.307	3C	1.787	(C,D)
M(PHI) / (P)	0.307	2C1	0.022	(A,B)
M(PHI) / (P)	0.307	1C	0.060	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.307	3A	1.098	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.307	1A	0.075	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.307	3B	2.278	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.307	1B	0.018	(A,B,C,D)
N(x) / (P/Rm)	0.307	3C	1.787	(A,B)
N(x) / (P/Rm)	0.307	4C	3.197	(C,D)



M(x)	/ (P)	0.307	1C1	0.047	(A,B)
M(x)	/ (P)	0.307	2C !	0.030	(C,D)
N(x)	/ (MC/(Rm**2 * Beta))	0.307	4A	2.268	(A,B,C,D)
M(x)	/ (MC/(Rm * Beta))	0.307	2A	0.032	(A,B,C,D)
N(x)	/ (ML/(Rm**2 * Beta))	0.307	4B	1.052	(A,B,C,D)
M(x)	/ (ML/(Rm * Beta))	0.307	2B	0.030	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-15.6	-15.6	15.6	15.6	0.0	0.0	0.0	0.0
Circ. Bend. ML		-19.7	19.7	19.7	-19.7	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-35.3	4.2	35.3	-4.2	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML		-7.2	-7.2	7.2	7.2	0.0	0.0	0.0	0.0
Long. Bend. ML		-32.6	32.6	32.6	-32.6	0.0	0.0	0.0	0.0
Tot. Long. Str.		-39.7	25.4	39.7	-25.4	0.0	0.0	0.0	0.0
Shear VC		0.0	0.0	-0.0	-0.0	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-4.3	-4.3	4.3	4.3
Shear MT		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tot. Shear		0.0	0.0	0.0	0.0	-4.3	-4.3	4.3	4.3
Str. Int.		39.7	25.4	39.7	25.4	8.7	8.7	8.7	8.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-11.1	-11.1	11.1	11.1	0.0	0.0	0.0	0.0
Circ. Q (SUS)		-44.7	44.7	44.7	-44.7	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-2.8	-2.8	2.8	2.8	0.0	0.0	0.0	0.0
Long. Q (SUS)		-72.2	72.2	72.2	-72.2	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	0.0	0.0	-6.2	-6.2	6.2	6.2
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		11.1	11.1	11.1	11.1	12.4	12.4	12.4	12.4

Pm+Pl+Q (Total)	75.1	69.4	75.1	69.4	12.4	12.4	12.4	12.4
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Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	12.38	207.00	Passed
Pm+Pl+Q (TOTAL)	75.08	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		-15.6	-15.6	15.6	15.6	0.0	0.0	0.0	0.0
Circ. Q (SUS)		-19.7	19.7	19.7	-19.7	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		-7.2	-7.2	7.2	7.2	0.0	0.0	0.0	0.0
Long. Q (SUS)		-32.6	32.6	32.6	-32.6	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.0	0.0	0.0	0.0	-4.3	-4.3	4.3	4.3
Shear Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		15.6	15.6	15.6	15.6	8.7	8.7	8.7	8.7
Pm+Pl+Q (Total)		39.7	25.4	39.7	25.4	8.7	8.7	8.7	8.7

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	15.56	207.00	Passed
Pm+Pl+Q (TOTAL)	39.75	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: SHELL-1

Section Properties : Single Angle L150X150X15

European Structural Steel Data

Overall Leg Length		1830.000	mm.
Effective Leg Length	Leglen	1600.000	mm.
Distance Leg Up Side of Vessel		250.000	mm.
Number of Legs	Nleg	4	
Cross Sectional Area for L150X150X15	Aleg	43.000	cm2
Section Inertia (strong axis)		897.996	cm4
Section Inertia (weak axis)		897.996	cm4
Section Modulus (strong axis)		83500.336	mm. ³
Section Modulus (weak axis)		83500.336	mm. ³
Radius of Gyration (strong axis)		45.700	mm.
Radius of Gyration (weak axis)		45.700	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		1532.6	Kg-m.
Total Weight Load at top of Legs	W	8033.2	Kgf
Total Shear force at top of Legs		1020.4	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	350.000	mm.
Pad Length along Vessel Axis	C22P	250.000	mm.
Pad Thickness	Tpad	20.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (1020) (1451/3646)$
 $= 406.07 \text{ Kg}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (78777/4 + (15028046 / (2 \cdot 713))) / 4300$
 $= 7.03 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (8033/4) / 43$
 $= 4.58 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	150.00	150.00	15.000
Distance to geometric centroid:	42.500	42.500	
Arm about YY:	35.000	40.000	
Arm about ZZ:	32.500	35.000	
Leg areas:	22.500	20.250	
Geometric inertia components YY:	279.84	631.54	
Geometric inertia components ZZ:	659.53	251.86	
Geometric inertias Iy & Iz:	911.38	911.38	
Product of inertia:	539.43		
Mohrs Radius:	539.43		
Average Inertia:	911.38		

QFACT =	1.0000	FBZ	=	0.16383	
Principal Axis Inertias (Z&W)			=	371.95	1450.8
Angle to Principal Axis			=	45.000	

Distances to extreme fibers CW & CZ = 106.07 45.962
 FOB from Eq 5-5 = 1.8261
 Bending allowables Fby & Fbz = 0.16383 0.16383

Shear Center Coordinates Wo & Zo: 48.393 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.77480 10.200
 AREA, LENGTH, Kw, Kz: 6.6650 62.992 1.0000 1.0000
 H, Few, Fez, Fej: 0.64414 383.73 98.378 129.31
 Fe computed from C4-1: 112.65

Initial (Kl/r)max, & (Kl/r)equiv = 54.402 50.839
 Final (Kl/r)max, & Cc = 54.402 127.18
 Fa based on Eq 4-1 = 0.12409

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	55.67	218.38
Strong Axis Bending :	32.94	218.38
Axial Compression :	7.03	165.42

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.448

AISC Unity Check : 0.448, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Ope Condition

Rectangular Attachment Parameter C11 212.132 mm.
 Rectangular Attachment Parameter C22 230.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	1231.000	mm.
Vessel Thickness	Tv	24.000	mm.
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Rectangular
Parameter C11	C11	212.13 mm.
Parameter C22	C22	230.95 mm.
Thickness of Reinforcing Pad	Tpad	20.000 mm.
Pad Parameter C11P	C11p	350.000 mm.
Pad Parameter C22P	C22p	250.000 mm.
Design Internal Pressure	Dp	40.998 bars
Include Pressure Thrust		No
Vessel Centerline Direction Cosine	Vx	0.000
Vessel Centerline Direction Cosine	Vy	1.000
Vessel Centerline Direction Cosine	Vz	0.000
Nozzle Centerline Direction Cosine	Nx	1.000
Nozzle Centerline Direction Cosine	Ny	0.000
Nozzle Centerline Direction Cosine	Nz	0.000



Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	2008.3	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	147.5	Kg-m.

Internal Pressure (SUS)	P	41.00	bars
Include Pressure Thrust		No	

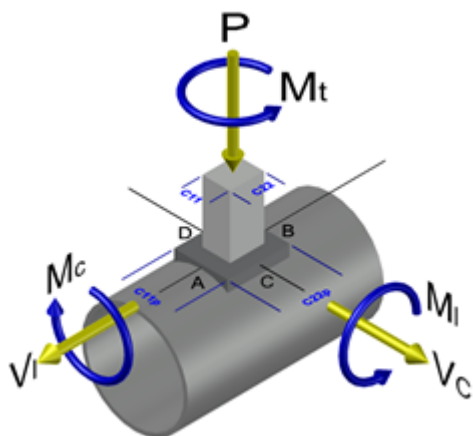
Global Force (OCC)	Fx	406.1	Kgf
Global Force (OCC)	Fy	1074.8	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	403.8	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	0.00	bars
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	2008.3	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-147.5	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 14.84

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.183	4C	2.537	(A,B)
N(PHI) / (P/Rm)	0.183	3C	2.168	(C,D)



M(PHI) / (P)	0.171	2C1	0.080	(A,B)
M(PHI) / (P)	0.171	1C	0.110	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.171	3A !	0.483	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.192	1A	0.094	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.176	3B !	1.574	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.180	1B	0.044	(A,B,C,D)
N(x) / (P/Rm)	0.177	3C	2.204	(A,B)
N(x) / (P/Rm)	0.177	4C	2.555	(C,D)
M(x) / (P)	0.178	1C1	0.110	(A,B)
M(x) / (P)	0.178	2C	0.076	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.171	4A !	0.711	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.214	2A	0.049	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.176	4B !	0.471	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.189	2B	0.066	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.7	0.7	-0.7	-0.7	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.8	-1.8	-1.8	1.8	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.5	-1.1	-2.5	1.1	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.6	-2.6	-2.6	2.6	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.8	-2.4	-2.8	2.4	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Str. Int.		2.8	2.4	2.8	2.4	2.0	2.0	2.0	2.0

Dimensionless Parameters used : Gamma = 27.30

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.233	4C	3.797	(A,B)
N(PHI) / (P/Rm)	0.233	3C	2.430	(C,D)
M(PHI) / (P)	0.259	2C1	0.029	(A,B)
M(PHI) / (P)	0.259	1C !	0.060	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.249	3A	1.158	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.286	1A	0.075	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.223	3B	2.878	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.236	1B	0.025	(A,B,C,D)
N(x) / (P/Rm)	0.257	3C	2.204	(A,B)

N(x)	/ (P/Rm)	0.257	4C	3.628	(C,D)
M(x)	/ (P)	0.242	1C1	0.063	(A,B)
M(x)	/ (P)	0.242	2C !	0.034	(C,D)
N(x)	/ (MC/(Rm**2 * Beta))	0.249	4A	2.148	(A,B,C,D)
M(x)	/ (MC/(Rm * Beta))	0.278	2A	0.033	(A,B,C,D)
N(x)	/ (ML/(Rm**2 * Beta))	0.223	4B	1.092	(A,B,C,D)
M(x)	/ (ML/(Rm * Beta))	0.233	2B	0.043	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.9	1.9	-1.9	-1.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.8	-2.8	-2.8	2.8	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		4.7	-0.9	-4.7	0.9	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.6	0.6	-0.6	-0.6	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		4.8	-4.8	-4.8	4.8	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		5.4	-4.1	-5.4	4.1	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Str. Int.		5.4	4.1	5.4	4.1	3.4	3.4	3.4	3.4

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	406.1	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	1074.8	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-403.8	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 14.84

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.4	-0.4	-0.4	-0.4	-0.3	-0.3	-0.3	-0.3
Circ. Bend. P		-1.0	1.0	-1.0	1.0	-1.4	1.4	-1.4	1.4
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.8	1.8	-1.8	-1.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		4.9	-4.9	-4.9	4.9	-0.0	-0.0	-0.0	-0.0

Tot. Circ. Str.	5.3	-2.5	-8.1	3.8	-1.7	1.1	-1.7	1.1
Long. Memb. P	-0.3	-0.3	-0.3	-0.3	-0.4	-0.4	-0.4	-0.4
Long. Bend. P	-1.4	1.4	-1.4	1.4	-1.0	1.0	-1.0	1.0
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	0.0
Long. Memb. ML	0.6	0.6	-0.6	-0.6	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	7.1	-7.1	-7.1	7.1	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	5.9	-5.4	-9.4	7.6	-1.4	0.6	-1.4	0.6
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Str. Int.	5.9	5.4	9.4	7.6	2.1	1.4	2.1	1.4

Dimensionless Parameters used : Gamma = 27.30
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-1.0	-1.0	-1.0	-1.0	-0.7	-0.7	-0.7	-0.7
Circ. Bend. P		-1.3	1.3	-1.3	1.3	-2.7	2.7	-2.7	2.7
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		5.1	5.1	-5.1	-5.1	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		7.6	-7.6	-7.6	7.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		10.4	-2.3	-15.1	2.8	-3.4	2.0	-3.4	2.0
Long. Memb. P		-0.6	-0.6	-0.6	-0.6	-1.0	-1.0	-1.0	-1.0
Long. Bend. P		-2.9	2.9	-2.9	2.9	-1.5	1.5	-1.5	1.5
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.8	1.8	-1.8	-1.8	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		13.1	-13.1	-13.1	13.1	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		11.4	-9.0	-18.3	13.5	-2.5	0.5	-2.5	0.5
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Str. Int.		11.4	9.0	18.3	13.5	4.0	2.5	4.0	2.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		56.8	60.9	56.8	60.9	56.8	60.9	56.8	60.9
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		56.8	60.9	56.8	60.9	56.8	60.9	56.8	60.9
Circ. Pl (SUS)		0.7	0.7	-0.7	-0.7	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		1.4	1.4	-2.2	-2.2	-0.3	-0.3	-0.3	-0.3
Circ. Pl(TOTAL)		2.1	2.1	-2.8	-2.8	-0.3	-0.3	-0.3	-0.3



Circ. Q (SUS)	1.8	-1.8	-1.8	1.8	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	3.9	-3.9	-6.0	6.0	-1.4	1.4	-1.4	1.4
Circ. Q (TOTAL)	5.7	-5.7	-7.8	7.8	-1.4	1.4	-1.4	1.4
Long. Pm (SUS)	28.4	28.4	28.4	28.4	28.4	28.4	28.4	28.4
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	28.4	28.4	28.4	28.4	28.4	28.4	28.4	28.4
Long. Pl (SUS)	0.2	0.2	-0.2	-0.2	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	0.3	0.3	-0.9	-0.9	-0.4	-0.4	-0.4	-0.4
Long. Pl(TOTAL)	0.5	0.5	-1.1	-1.1	-0.4	-0.4	-0.4	-0.4
Long. Q (SUS)	2.6	-2.6	-2.6	2.6	0.0	0.0	-0.0	0.0
Long. Q (OCC)	5.7	-5.7	-8.5	8.5	-1.0	1.0	-1.0	1.0
Long. Q (TOTAL)	8.2	-8.2	-11.1	11.1	-1.0	1.0	-1.0	1.0
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	56.8	60.9	56.8	60.9	56.8	60.9	56.8	60.9
Pm (SUS+OCC)	56.8	60.9	56.8	60.9	56.8	60.9	56.8	60.9
Pm+Pl (SUS)	57.5	61.6	56.1	60.2	56.8	60.9	56.8	60.9
Pm+Pl (SUS+OCC)	58.9	63.0	54.0	58.1	56.6	60.7	56.6	60.7
Pm+Pl+Q (Total)	64.6	57.3	46.2	65.9	55.2	62.1	55.2	62.1

Vessel Stress Summation Comparison (N/mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	60.90	138.00	Passed
Pm (SUS+OCC)	60.90	165.60	Passed
Pm+Pl (SUS)	61.56	207.00	Passed
Pm+Pl (SUS+OCC)	62.99	248.40	Passed
Pm+Pl+Q (TOTAL)	65.86	468.82	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N/mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		107.9	112.0	107.9	112.0	107.9	112.0	107.9	112.0
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		107.9	112.0	107.9	112.0	107.9	112.0	107.9	112.0
Circ. Pl (SUS)		1.9	1.9	-1.9	-1.9	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		4.1	4.1	-6.1	-6.1	-0.7	-0.7	-0.7	-0.7
Circ. Pl(TOTAL)		5.9	5.9	-8.0	-8.0	-0.7	-0.7	-0.7	-0.7
Circ. Q (SUS)		2.8	-2.8	-2.8	2.8	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		6.3	-6.3	-8.9	8.9	-2.7	2.7	-2.7	2.7
Circ. Q (TOTAL)		9.1	-9.1	-11.7	11.7	-2.7	2.7	-2.7	2.7

Long. Pm (SUS)	53.9	53.9	53.9	53.9	53.9	53.9	53.9	53.9
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	53.9	53.9	53.9	53.9	53.9	53.9	53.9	53.9
Long. P1 (SUS)	0.6	0.6	-0.6	-0.6	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	1.2	1.2	-2.4	-2.4	-1.0	-1.0	-1.0	-1.0
Long. P1(TOTAL)	1.8	1.8	-3.0	-3.0	-1.0	-1.0	-1.0	-1.0
Long. Q (SUS)	4.8	-4.8	-4.8	4.8	0.0	0.0	-0.0	0.0
Long. Q (OCC)	10.2	-10.2	-15.9	15.9	-1.5	1.5	-1.5	1.5
Long. Q (TOTAL)	15.0	-15.0	-20.7	20.7	-1.5	1.5	-1.5	1.5
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-2.6	-2.6	2.6	2.6
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	107.9	112.0	107.9	112.0	107.9	112.0	107.9	112.0
Pm (SUS+OCC)	107.9	112.0	107.9	112.0	107.9	112.0	107.9	112.0
Pm+P1 (SUS)	109.7	113.8	106.0	110.1	107.9	112.0	107.9	112.0
Pm+P1 (SUS+OCC)	113.8	117.9	99.9	104.0	107.3	111.4	107.3	111.4
Pm+P1+Q (Total)	122.9	108.8	88.2	115.7	104.6	114.1	104.6	114.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	111.99	138.00	Passed
Pm (SUS+OCC)	111.99	165.60	Passed
Pm+P1 (SUS)	113.85	207.00	Passed
Pm+P1 (SUS+OCC)	117.90	248.40	Passed
Pm+P1+Q (TOTAL)	122.92	468.82	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 N./mm ²
Baseplate Length	B 350.0000 mm.
Baseplate Width	D 350.0000 mm.
Baseplate Thickness	BTHK 26.0000 mm.
Leg Dimension Along Baseplate Length	d 149.9997 mm.
Leg Dimension Along Baseplate Width	b 149.9997 mm.
Bolt Circle Diameter	BCD 1251.0000 mm.
Bolt Material	SA-307 A
Bolt Allowable Tensile Stress	STBA 138.00 N./mm ²
Bolt Allowable Shear Stress	SBShear 28.96 N./mm ²
Anchor Bolt Nominal Diameter	BOD 30.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB 1
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 3.0000 mm.
Ultimate 28-day Concrete Strength	FCPRIME 20.685 N./mm ²

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:
 = Shear Force/(Bolt Area • Number of Bolts)
 = 1020/(4.04 • 4)



= 6.2 N./mm². Must be less than 29.0 N./mm².

Plate thickness required (5/8"; 16mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Computing the required bolt area based on a standard method not including the baseplate's interaction with the foundation.

Calculation of Load per Bolt, Wind Empty [W,wind]:
= ((4 • Moment/BCD) - Weight)/Num Bolts
= ((4 • 267/1251) - 7340)/4
= -1621.736 Kgf

Calculation of Load per Bolt, Seismic Operating [W,seismic]:
= ((4 • Moment/BCD) - Weight)/Num Bolts
= ((4 • 1533/1251) - 8377)/4
= -869.189 Kgf

Note: The bolts are not loaded in tension.

Review notes about nozzle loadings and supports in the Warnings report.

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: SHELL-1

Section Properties : Single Angle L150X150X15

European Structural Steel Data

Overall Leg Length		1830.000	mm.
Effective Leg Length	Leglen	1600.000	mm.
Distance Leg Up Side of Vessel		250.000	mm.
Number of Legs	Nleg	4	
Cross Sectional Area for L150X150X15	Aleg	43.000	cm ²
Section Inertia (strong axis)		897.996	cm ⁴
Section Inertia (weak axis)		897.996	cm ⁴
Section Modulus (strong axis)		83500.336	mm. ³
Section Modulus (weak axis)		83500.336	mm. ³
Radius of Gyration (strong axis)		45.700	mm.
Radius of Gyration (weak axis)		45.700	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		165.3	Kg-m.
Total Weight Load at top of Legs	W	9933.0	Kgf
Total Shear force at top of Legs		111.0	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	350.000	mm.
Pad Length along Vessel Axis	C22P	250.000	mm.
Pad Thickness	Tpad	20.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (111) (1451/3646)$
 $= 44.19 \text{ Kgf}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (97408/4 + (1621359 / (2 \cdot 713))) / 4300$
 $= 5.93 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (9933/4) / 43$
 $= 5.66 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	150.00	150.00	15.000
Distance to geometric centroid:	42.500	42.500	
Arm about YY:	35.000	40.000	
Arm about ZZ:	32.500	35.000	
Leg areas:	22.500	20.250	
Geometric inertia components YY:	279.84	631.54	
Geometric inertia components ZZ:	659.53	251.86	
Geometric inertias Iy & Iz:	911.38	911.38	
Product of inertia:	539.43		
Mohrs Radius:	539.43		
Average Inertia:	911.38		

QFACT =	1.0000	FBZ	=	0.16383	
Principal Axis Inertias (Z&W)			=	371.95	1450.8
Angle to Principal Axis			=	45.000	



Distances to extreme fibers CW & CZ = 106.07 45.962
 FOB from Eq 5-5 = 1.8261
 Bending allowables Fby & Fbz = 0.16383 0.16383
 Shear Center Coordinates Wo & Zo: 48.393 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.77480 10.200
 AREA, LENGTH, Kw, Kz: 6.6650 62.992 1.0000 1.0000
 H, Few, Fez, Fej: 0.64414 383.73 98.378 129.31
 Fe computed from C4-1: 112.65
 Initial (Kl/r)max, & (Kl/r)equiv = 54.402 50.839
 Final (Kl/r)max, & Cc = 54.402 127.18
 Fa based on Eq 4-1 = 0.12409

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	6.06	218.38
Strong Axis Bending :	3.58	218.38
Axial Compression :	5.93	165.42
UNITY CHECKS ARE: H1-1 0.000		
	H1-2 0.000	
	H1-3 0.080	

AISC Unity Check : 0.080, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Test Condition

Rectangular Attachment Parameter	C11	212.132 mm.
Rectangular Attachment Parameter	C22	230.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	1.0000 mm.
Vessel Diameter	Dv	1231.000 mm.
Vessel Thickness	Tv	24.000 mm.
Design Temperature	T1	130.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	234.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	234.00 N./mm ²
Attachment Type	Type	Rectangular
Parameter C11	C11	212.13 mm.
Parameter C22	C22	230.95 mm.
Thickness of Reinforcing Pad	Tpad	20.000 mm.
Pad Parameter C11P	C11p	350.000 mm.
Pad Parameter C22P	C22p	250.000 mm.
Design Internal Pressure	Dp	40.998 bars
Include Pressure Thrust		No
Vessel Centerline Direction Cosine	Vx	0.000
Vessel Centerline Direction Cosine	Vy	1.000
Vessel Centerline Direction Cosine	Vz	0.000
Nozzle Centerline Direction Cosine	Nx	1.000
Nozzle Centerline Direction Cosine	Ny	0.000
Nozzle Centerline Direction Cosine	Nz	0.000
Global Force (SUS)	Fx	0.0 Kg
Global Force (SUS)	Fy	2483.2 Kg



Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	182.3	Kg-m.

Internal Pressure (SUS)	P	41.00	bars
Include Pressure Thrust		No	

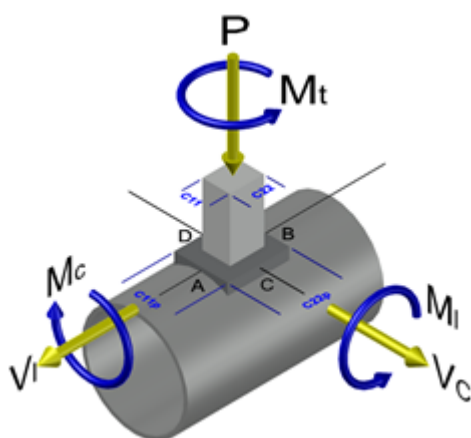
Global Force (OCC)	Fx	44.2	Kgf
Global Force (OCC)	Fy	116.0	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	43.9	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	12.54	bars
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	2483.2	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-182.3	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 14.84

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.183	4C	2.537	(A,B)
N(PHI) / (P/Rm)	0.183	3C	2.168	(C,D)
M(PHI) / (P)	0.171	2C1	0.080	(A,B)
M(PHI) / (P)	0.171	1C	0.110	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.171	3A	0.483	(A,B,C,D)



M(PHI) / (MC/(Rm * Beta))	0.192	1A	0.094	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.176	3B !	1.574	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.180	1B	0.044	(A,B,C,D)
N(x) / (P/Rm)	0.177	3C	2.204	(A,B)
N(x) / (P/Rm)	0.177	4C	2.555	(C,D)
M(x) / (P)	0.178	1C1	0.110	(A,B)
M(x) / (P)	0.178	2C	0.076	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.171	4A !	0.711	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.214	2A	0.049	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.176	4B !	0.471	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.189	2B	0.066	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.8	0.8	-0.8	-0.8	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.2	-2.2	-2.2	2.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		3.0	-1.4	-3.0	1.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		3.2	-3.2	-3.2	3.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		3.5	-2.9	-3.5	2.9	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Str. Int.		3.5	2.9	3.5	2.9	2.5	2.5	2.5	2.5

Dimensionless Parameters used : Gamma = 27.30

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.233	4C	3.797	(A,B)
N(PHI) / (P/Rm)	0.233	3C	2.430	(C,D)
M(PHI) / (P)	0.259	2C1	0.029	(A,B)
M(PHI) / (P)	0.259	1C !	0.060	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.249	3A	1.158	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.286	1A	0.075	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.223	3B	2.878	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.236	1B	0.025	(A,B,C,D)
N(x) / (P/Rm)	0.257	3C	2.204	(A,B)
N(x) / (P/Rm)	0.257	4C	3.628	(C,D)
M(x) / (P)	0.242	1C1	0.063	(A,B)
M(x) / (P)	0.242	2C !	0.034	(C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

WRC 107 Stress Calculation for OCCasional loads:

Dimensionless Parameters used : $\Gamma = 14.84$
Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

[illegible]



Long. Bend. P	-0.2	0.2	-0.2	0.2	-0.1	0.1	-0.1	0.1
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML	0.1	0.1	-0.1	-0.1	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.8	-0.8	-0.8	0.8	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	0.6	-0.6	-1.0	0.8	-0.1	0.1	-0.1	0.1
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Str. Int.	0.6	0.6	1.0	0.8	0.2	0.2	0.2	0.2

Dimensionless Parameters used : Gamma = 27.30
 Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Circ. Bend. P		-0.1	0.1	-0.1	0.1	-0.3	0.3	-0.3	0.3
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.6	0.6	-0.6	-0.6	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.8	-0.8	-0.8	0.8	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		1.1	-0.2	-1.6	0.3	-0.4	0.2	-0.4	0.2
Long. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Long. Bend. P		-0.3	0.3	-0.3	0.3	-0.2	0.2	-0.2	0.2
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		1.4	-1.4	-1.4	1.4	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		1.2	-1.0	-2.0	1.5	-0.3	0.1	-0.3	0.1
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Str. Int.		1.2	1.0	2.0	1.5	0.4	0.3	0.4	0.3

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		56.8	60.9	56.8	60.9	56.8	60.9	56.8	60.9
Circ. Pm (OCC)		17.4	18.6	17.4	18.6	17.4	18.6	17.4	18.6
Circ. Pm(TOTAL)		74.2	79.5	74.2	79.5	74.2	79.5	74.2	79.5
Circ. Pl (SUS)		0.8	0.8	-0.8	-0.8	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Circ. Pl(TOTAL)		1.0	1.0	-1.0	-1.0	-0.0	-0.0	-0.0	-0.0
Circ. Q (SUS)		2.2	-2.2	-2.2	2.2	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		0.4	-0.4	-0.6	0.6	-0.2	0.2	-0.2	0.2
Circ. Q (TOTAL)		2.7	-2.7	-2.9	2.9	-0.2	0.2	-0.2	0.2



Long. Pm (SUS)	28.4	28.4	28.4	28.4	28.4	28.4	28.4	28.4
Long. Pm (OCC)	8.7	8.7	8.7	8.7	8.7	8.7	8.7	8.7
Long. Pm(TOTAL)	37.1	37.1	37.1	37.1	37.1	37.1	37.1	37.1
Long. Pl (SUS)	0.3	0.3	-0.3	-0.3	0.0	0.0	-0.0	-0.0
Long. Pl (OCC)	0.0	0.0	-0.1	-0.1	-0.0	-0.0	-0.0	-0.0
Long. Pl(TOTAL)	0.3	0.3	-0.4	-0.4	-0.0	-0.0	-0.0	-0.0
Long. Q (SUS)	3.2	-3.2	-3.2	3.2	0.0	0.0	-0.0	0.0
Long. Q (OCC)	0.6	-0.6	-0.9	0.9	-0.1	0.1	-0.1	0.1
Long. Q (TOTAL)	3.8	-3.8	-4.1	4.1	-0.1	0.1	-0.1	0.1
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear Pl (OCC)	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear Pl(TOTAL)	0.0	0.0	0.0	0.0	-1.3	-1.3	1.3	1.3
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	56.8	60.9	56.8	60.9	56.8	60.9	56.8	60.9
Pm (SUS+OCC)	74.2	79.5	74.2	79.5	74.2	79.5	74.2	79.5
Pm+Pl (SUS)	57.6	61.7	56.0	60.1	56.9	60.9	56.9	60.9
Pm+Pl (SUS+OCC)	75.1	80.5	73.1	78.5	74.2	79.5	74.2	79.5
Pm+Pl+Q (Total)	77.8	77.8	70.2	81.4	74.0	79.7	74.0	79.7

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	60.90	234.00	Passed
Pm (SUS+OCC)	79.53	280.80	Passed
Pm+P1 (SUS)	61.71	351.00	Passed
Pm+P1 (SUS+OCC)	80.50	421.20	Passed
Pm+P1+0 (TOTAL)	81.36	702.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The P_m+P_I+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)

[illegible]



Long. P1 (SUS)	0.8	0.8	-0.8	-0.8	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.1	0.1	-0.3	-0.3	-0.1	-0.1	-0.1	-0.1
Long. P1(TOTAL)	0.9	0.9	-1.1	-1.1	-0.1	-0.1	-0.1	-0.1
Long. Q (SUS)	5.9	-5.9	-5.9	5.9	0.0	0.0	-0.0	0.0
Long. Q (OCC)	1.1	-1.1	-1.7	1.7	-0.2	0.2	-0.2	0.2
Long. Q (TOTAL)	7.0	-7.0	-7.6	7.6	-0.2	0.2	-0.2	0.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	107.9	112.0	107.9	112.0	107.9	112.0	107.9	112.0
Pm (SUS+OCC)	140.9	146.2	140.9	146.2	140.9	146.2	140.9	146.2
Pm+P1 (SUS)	110.2	114.3	105.6	109.7	108.0	112.1	108.0	112.1
Pm+P1 (SUS+OCC)	143.6	149.0	137.9	143.3	140.9	146.2	140.9	146.2
Pm+P1+Q (Total)	147.8	144.8	133.5	147.7	140.6	146.5	140.6	146.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	111.99	234.00	Passed
Pm (SUS+OCC)	146.24	280.80	Passed
Pm+P1 (SUS)	114.29	351.00	Passed
Pm+P1 (SUS+OCC)	148.98	421.20	Passed
Pm+P1+Q (TOTAL)	147.76	702.00	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material		SA-516 70
Baseplate Allowable Stress	SBA	137.90 N./mm ²
Baseplate Length	B	350.0000 mm.
Baseplate Width	D	350.0000 mm.
Baseplate Thickness	BTHK	26.0000 mm.
Leg Dimension Along Baseplate Length	d	149.9997 mm.
Leg Dimension Along Baseplate Width	b	149.9997 mm.
Bolt Circle Diameter	BCD	1251.0000 mm.
Bolt Material		SA-307 A
Bolt Allowable Tensile Stress	STBA	138.00 N./mm ²
Bolt Allowable Shear Stress	SBShear	28.96 N./mm ²
Anchor Bolt Nominal Diameter	BOD	30.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB	1
Total Number of Anchors Bolt per Leg	NBT	1
Bolt Corrosion Allowance	bca	3.0000 mm.
Ultimate 28-day Concrete Strength	FCPRIME	20.685 N./mm ²

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [rb]:

= Shear Force/(Bolt Area • Number of Bolts)

= 1020/(4.04 • 4)

= 6.2 N./mm². Must be less than 29.0 N./mm².

Plate thickness required (5/8"; 16mm) per D. Moss



Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N3-2"

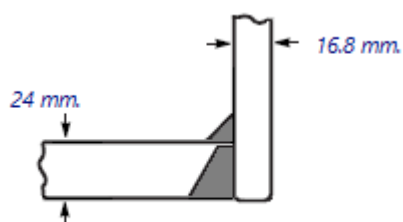
From: 10

Pressure for Reinforcement Calculations	P	41.005	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1231.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	220.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	19.5000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		750.0	mm.
Centerline Distance to Tangent		625.0	mm.

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 41 \cdot 1233 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 41)$
 $= 16.5110$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([41 / (118 \cdot 1)] - 1))$
 $= 0.9336$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	130.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	65.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	39.4100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

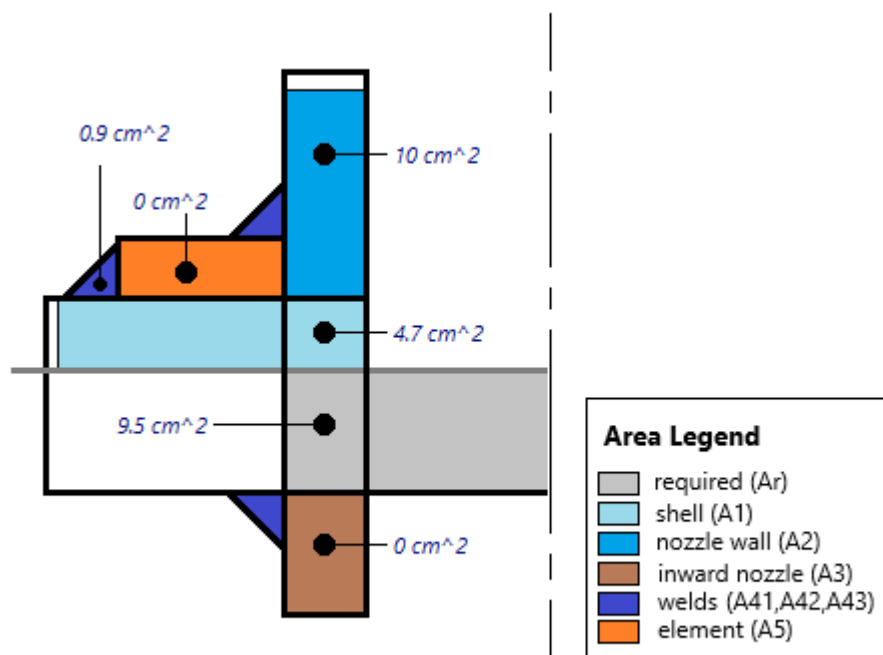


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9.472	NA	NA
Area in Shell	A1	4.734	NA	NA
Area in Nozzle Wall	A2	9.995	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	15.584	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (52.8 \cdot 16.5 \cdot 1 + 2 \cdot 15.8 \cdot 16.5 \cdot 1 \cdot (1 - 0.86))$
 $= 9.472 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5(1 \cdot 23 - 1 \cdot 16.5) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 16.5) \cdot (1 - 0.86)$
 $= 4.734 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 39.4)(15.8 - 0.93)0.86$
 $= 9.995 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 0.855 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.9336 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.3344 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.3344 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(19.3, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.93, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 18.2,	Allowable	: 118.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 276.8 N./mm2	Passed
Occasional	: 2.6,	Allowable	: 156.9 N./mm2	Passed
Shear	: 11.8,	Allowable	: 82.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 16.8, tr = 0.93, c = 1 mm., E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.059$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -13 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $41.00/102.00 = 0.402$

Weld Size Calculations, Description: N3-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9.47 - 4.73 + 2 \cdot 15.8 \cdot 0.86 \cdot (1 \cdot 23 - 16.5)) \cdot 138)$
 $= 9128.99 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (10 + 0 + 0.86 - 0 \cdot 0.86) \cdot 138$
 $= 15268.07 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (10 + 0 + 0.86 + (6.2)) \cdot 138$
 $= 23993.16 \text{ Kg}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (10 + 0 + 0.86 + 0 + (6.2)) \cdot 138$
 $= 23993.16 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 118$
 $= 7810. \text{ Kg}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 118$
 $= 14300. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 84.3 \cdot (19.5 - 0) \cdot 0.74 \cdot 138$
 $= 26897. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (7810 + 14300) = 22109 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (7810 + 0 + 26897 + 0) = 34707 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (7810 + 26897 + 0) = 34707 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 22109	Kgf, must exceed W = 9128 Kg	or W1 = 15268 Kg
Path 2-2 = 34706	Kgf, must exceed W = 9128 Kg	or W2 = 23993 Kg
Path 3-3 = 34706	Kgf, must exceed W = 9128 Kg	or W3 = 23993 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 41 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.7890 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 244.7890 mm.

Input Echo, WRC107/537 Item 1, Description: N3-2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	2215.800	mm.
Vessel Thickness	Tv	24.000	mm.
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.0000	mm.
Nozzle Diameter	Dn	50.800	mm.
Nozzle Thickness	Tn	16.764	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Design Internal Pressure	Dp	41.005	bars
Include Pressure Thrust		No	

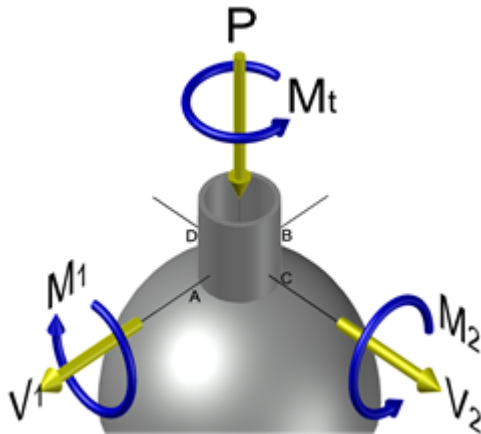
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	458.9	Kgf
Longitudinal Shear	(SUS)	(V1) V1	331.4	Kgf
Circumferential Shear	(SUS)	(Vc) V2	331.4	Kgf
Circumferential Moment	(SUS)	(Mc) M1	51.0	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	51.0	Kg-m.
Torsional Moment	(SUS)	Mt	58.6	Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(1120.4 (24.0 - 1.0))}$$

$$= 614.070 \text{ mm.}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	458.9	Kgf
Circumferential Shear	(VC) V2	331.4	Kgf
Longitudinal Shear	(VL) V1	331.4	Kgf
Circumferential Moment	(MC) M1	51.0	Kg-m.
Longitudinal Moment	(ML) M2	51.0	Kg-m.
Torsional Moment	MT	58.6	Kg-m.

Dimensionless Param: U = 0.26 TAU = 5.00 (2.17) RHO = 1.46

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.11468	(A,B,C,D)
$M(x) / P$	SP 2	0.12223	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.29435	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.41828	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.29435	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.41828	(A,B,C,D)
$N(y) * T / P$	SP 2	0.20390	(A,B,C,D)
$M(y) / P$	SP 2	0.07277	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.04591	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.32768	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.04591	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.32768	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Stress Intensity Values at



Type of Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0
Rad.	Bend. P	-6.2	6.2	-6.2	6.2	-6.2	6.2	-6.2	6.2
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-14.8	14.8	14.8	-14.8
Rad.	Memb. ML	-1.7	-1.7	1.7	1.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-14.8	14.8	14.8	-14.8	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-23.7	18.3	9.3	-7.8	-23.7	18.3	9.3	-7.8
Tang.	Memb. P	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7	-1.7
Tang.	Bend. P	-3.7	3.7	-3.7	3.7	-3.7	3.7	-3.7	3.7
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-11.6	11.6	11.6	-11.6
Tang.	Memb. ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-11.6	11.6	11.6	-11.6	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-17.3	13.3	6.4	-9.3	-17.3	13.3	6.4	-9.3
	Shear VC	1.1	1.1	-1.1	-1.1	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
	Shear MT	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
	Tot. Shear	3.3	3.3	1.2	1.2	1.2	1.2	3.3	3.3
	Str. Int.	25.1	19.9	9.7	10.0	23.9	18.6	11.4	11.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9
Rad.	Pl (SUS)	-2.7	-2.7	0.8	0.8	-2.7	-2.7	0.8	0.8
Rad.	Q (SUS)	-21.0	21.0	8.5	-8.5	-21.0	21.0	8.5	-8.5
Long.	Pm (SUS)	98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9
Long.	Pl (SUS)	-2.0	-2.0	-1.5	-1.5	-2.0	-2.0	-1.5	-1.5
Long.	Q (SUS)	-15.3	15.3	7.9	-7.9	-15.3	15.3	7.9	-7.9
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.1	1.1	-1.1	-1.1	-1.1	-1.1	1.1	1.1
Shear	Q (SUS)	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
	Pm (SUS)	98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9
	Pm+Pl (SUS)	97.6	97.6	100.1	100.1	97.6	97.6	100.1	100.1
	Pm+Pl+Q (Total)	83.0	118.8	108.6	91.7	81.8	117.4	110.3	93.7

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	98.88	138.00	Passed
Pm+Pl (SUS)	100.07	207.00	Passed
Pm+Pl+Q (TOTAL)	118.82	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N1-4"

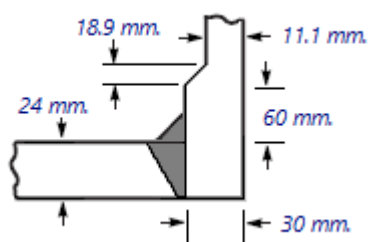
From: 20

Pressure for Reinforcement Calculations	P	41.005	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1231.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1250.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	60.0000	mm.
Height of Beveled Transition	L'	18.8700	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	280.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N1-4"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 4.500 in.
 Actual Thickness Used in Calculation 0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 41 \cdot 616 / (138 \cdot 1 - 0.6 \cdot 41)$
 $= 18.6520$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 41 \cdot 57.1 / (138 \cdot 1 + 0.4 \cdot 41)$
 $= 1.6783$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 18.9 / 0.5773$
 $= 32.6921$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

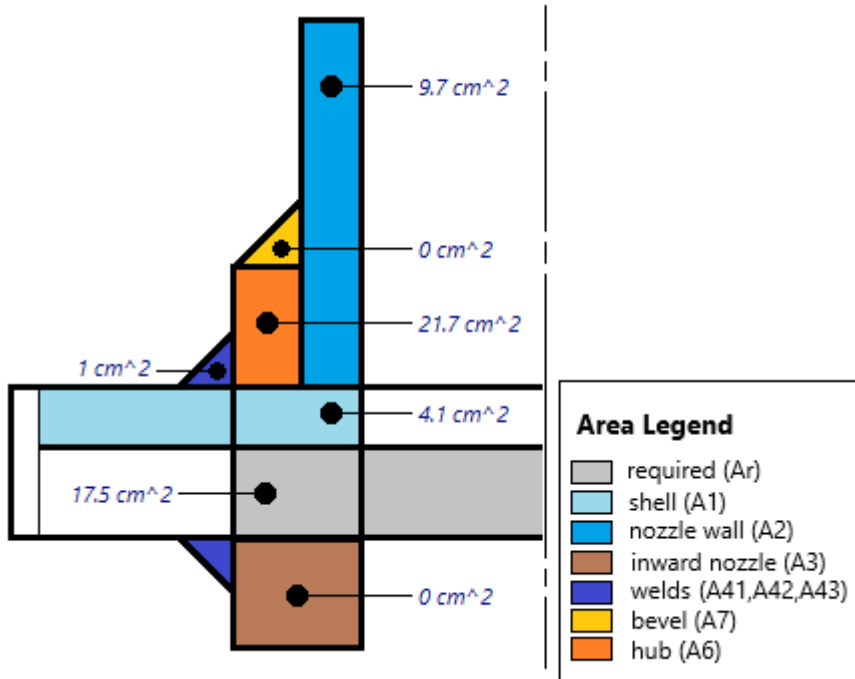
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	188.0992	mm.
Parallel to Vessel Wall, opening length	d	94.0496	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	57.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	17.542	NA	NA
Area in Shell	A1	4.089	NA	NA
Area in Nozzle Wall	A2	9.714	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	21.706	NA	NA
TOTAL AREA AVAILABLE	Atot	36.509	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (94 \cdot 18.7 \cdot 1 + 2 \cdot 29 \cdot 18.7 \cdot 1 \cdot (1 - 1))$
 $= 17.542 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 94 (1 \cdot 23 - 1 \cdot 18.7) - 2 \cdot 29 (1 \cdot 23 - 1 \cdot 18.7) \cdot (1 - 1)$
 $= 4.089 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 57.5) (10.1 - 1.68) 1$
 $= 9.714 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(57.5, 280, 60)) \cdot (30 - 11.1) \cdot 1$$

$$= 21.706 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 2.6783 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.6520 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.6520 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 6.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.26, \max(19.7, 2.5)]$$

$$= 6.2578 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.68, 6.26)$$

$$= 6.2578 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	46.9,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	298.1 N./mm ²	Passed
Occasional	:	8.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	22.5,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $tg = 9.73$, $tr = 1.68$, $c = 1 \text{ mm.}$, $E* = 1$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.19$, Temp. Reduction = $78 \text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $-8 \text{ }^{\circ}\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^{\circ}\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, $tg = 24$, $tr = 18.7$, $c = 1 \text{ mm.}$, $E* = 1$
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.81$, Temp. Reduction = $11 \text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $-2 \text{ }^{\circ}\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-13 \text{ }^{\circ}\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-13 \text{ }^{\circ}\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^{\circ}\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) $-48 \text{ }^{\circ}\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $41.00/102.00 = 0.402$

Weld Size Calculations, Description: N1-4"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (17.5 - 4.09 + 2 \cdot 29 \cdot 1 \cdot (1 \cdot 23 - 18.7)) \cdot 138)$
 $= 22479.55 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (31.4 + 1 - 0 \cdot 1) \cdot 138$
 $= 45619.96 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (31.4 + 0 + 1 + (13.3)) \cdot 138$
 $= 64391.43 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (31.4 + 0 + 1 + 0 + (13.3)) \cdot 138$
 $= 64391.43 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 138$
 $= 12380. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 52.1) \cdot (30 - 1) \cdot 0.7 \cdot 138$
 $= 46743. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 114 \cdot (20 - 0) \cdot 0.74 \cdot 138$
 $= 37392. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (12380 + 46743) = 59123 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (12380 + 0 + 37392 + 0) = 49772 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (12380 + 37392 + 0) = 49772 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 59123	Kgf, must exceed W = 22479 Kgf	or W1 = 45619 Kgf
Path 2-2 = 49771	Kgf, must exceed W = 22479 Kgf	or W2 = 64391 Kgf
Path 3-3 = 49771	Kgf, must exceed W = 22479 Kgf	or W3 = 64391 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 41 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 2.6590 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 306.6590 mm.

Input Echo, WRC297 Item 1, Description: N1-4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1231.000	mm.
Shell Thickness	24.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.0000	mm.
Nozzle Diameter	152.050	mm.
Nozzle Thickness	30.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	41.00	bars
Radial Load	P	917.81	Kgf
Circumferential Shear	Vc	662.86	Kgf
Longitudinal Shear	Vl	662.86	Kgf
Circumferential Moment	Mc	203.88	Kg-m.
Longitudinal Moment	Ml	203.88	Kg-m.
Torsional Moment	Mt	234.46	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.068) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 152.05 + 2 \cdot 1.65 \cdot \sqrt{628.0(24.0 - 1.0)}$$

$$= 548.654 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1256.000 mm.
Nozzle Outside Diameter	(d) =	152.050 mm.
Vessel Thickness used	(T) =	23.000 mm.
Nozzle Thickness used	(t) =	29.000 mm.
T / t	=	0.793
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.895$$

Nr/P	=	0.125
Mr/P	=	0.218



M0/P = 0.070
 N0/P = 0.132
 MrD/Mc = 0.293
 NrDL/Mc = 0.122
 M0d/Mc = 0.105
 N0DL/Mc = -0.019
 MrD/M1 = 0.210
 NrDL/M1 = 0.106
 M0D/M1 = 0.075
 N0DL/M1 = 0.106

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-7	7	-7	7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	107	112	107	112
Outplane Stress Summary	84	125	112	108

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-22	22	-22	22
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-31	31	31	-31
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	-4	102	63	45

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	89	126	112	108

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-2	-2	-2	-2
Outplane Bending (P)	-7	7	-7	7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-15	15	15	-15
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	53	53	53	53
------------------------	----	----	----	----

Outplane Stress Summary	29	74	59	42
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	-2	-2	-2	-2
Inplane Bending (P)	-22	22	-22	22
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-43	43	43	-43
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	107	112	107	112

Inplane Stress Summary	36	172	130	91
------------------------	----	-----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	37	172	130	91
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	11	6	11

Hoop Stress Summary	1	6	6	11
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-9	9	-9	9
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-14	14	14	-14
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-29	20	13	4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	31	20	13	11
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	-2	-2	-2	-2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	11	6	11

Hoop Stress Summary	4	9	3	8
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-9	9	-9	9
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-21	21	21	-21
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-36	27	20	-2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	40	27	21	12
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	107	112	107	112	107	112	107	112
Circ. Pl (SUS)	-4	-4	0	0	-5	-5	1	1
Circ. Q (SUS)	-18	18	4	-4	-65	65	21	-21



Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-4	-4	0	0	-2	-2	-2	-2
Long. Q (SUS)	-53	53	9	-9	-22	22	8	-8
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	107.0	112.0	107.0	112.0	107.0	112.0	107.0	112.0
Pm+Pl (SUS)	103.0	108.0	107.0	112.0	102.0	107.0	108.0	113.0
Pm+Pl+Q (Total)	89.2	126.4	111.0	108.0	37.1	172.0	129.1	92.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	112.00	138.00	Passed
Pm+Pl (SUS)	113.02	207.00	Passed
Pm+Pl+Q (TOTAL)	172.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	11	6	11	6	11	6	11
Circ. Pl (SUS)	-4	-4	0	0	-2	-2	-2	-2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-23	23	5	-5	-30	30	12	-12
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	6.0	11.0	6.0	11.0	6.0	11.0	6.0	11.0
Pm+Pl (SUS)	5.0	10.0	9.0	11.0	7.0	12.0	9.0	9.0
Pm+Pl+Q (Total)	28.3	20.3	14.5	11.5	37.2	27.2	21.2	12.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	138.00	Passed
Pm+Pl (SUS)	12.00	207.00	Passed
Pm+Pl+Q (TOTAL)	37.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: K1-2"

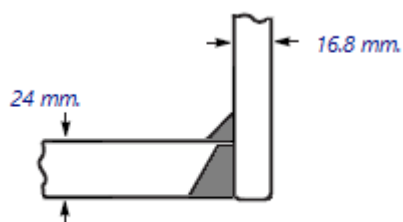
From: 20

Pressure for Reinforcement Calculations	P	41.005	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1231.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		900.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	240.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 41 \cdot 616 / (138 \cdot 1 - 0.6 \cdot 41)$
 $= 18.6520$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([41 / (115 \cdot 1)] - 1))$
 $= 0.9584$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 130.3280 mm.
 Parallel to Vessel Wall Rn+tn+t 65.1640 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.4100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

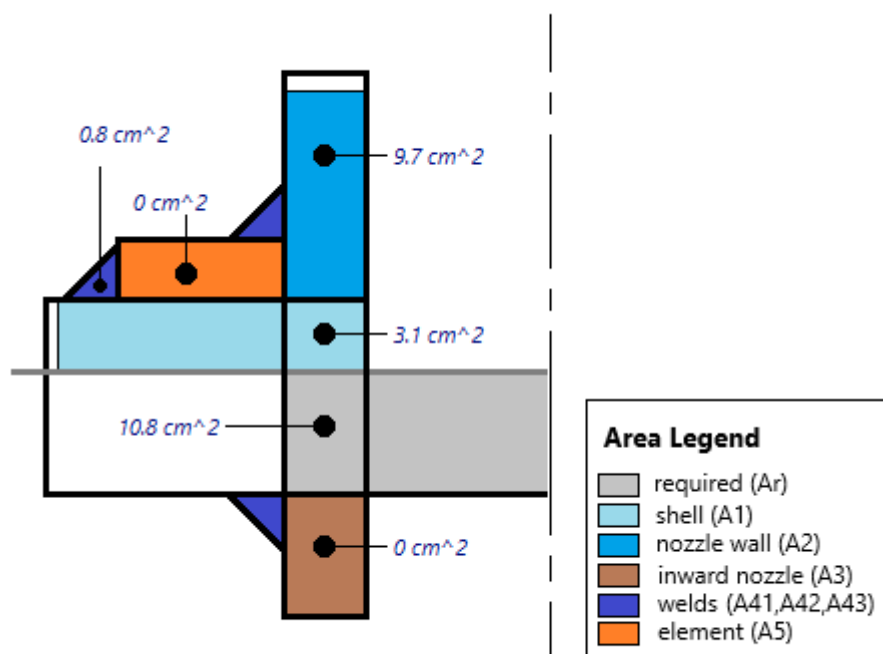


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10.828	NA	NA
Area in Shell	A1	3.142	NA	NA
Area in Nozzle Wall	A2	9.725	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA

Area in Welds	A41+A42+A43	0.833	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	13.700	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (52.8 \cdot 18.7 \cdot 1 + 2 \cdot 15.8 \cdot 18.7 \cdot 1 \cdot (1 - 0.83))$
 $= 10.828 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5(1 \cdot 23 - 1 \cdot 18.7) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 18.7) \cdot (1 - 0.83)$
 $= 3.142 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tnp)(tn - trn)fr2$
 $= (2 \cdot 39.4)(15.8 - 0.96)0.83$
 $= 9.725 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.9584 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.6520 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.6520 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(19.7, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.96, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (10.8 - 3.14 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 23 - 18.7)) \cdot 138) \\
 &= 12423.09 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v \\
 &= (9.72 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138 \\
 &= 14856.99 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\
 &= (9.72 + 0 + 0.83 + (6.04)) \cdot 138 \\
 &= 23360.25 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (9.72 + 0 + 0.83 + 0 + (6.04)) \cdot 138 \\
 &= 23360.25 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115 \\
 &= 7611. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115 \\
 &= 13936. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{g n v i} \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 84.3 \cdot (20 - 0) \cdot 0.74 \cdot 138 \\
 &= 27587. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7611 + 13936) = 21547 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (7611 + 0 + 27587 + 0) = 35198 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (7611 + 27587 + 0) = 35198 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 12423 Kgf	or W1 = 14856 Kgf
Path 2-2 = 35198	Kgf, must exceed W = 12423 Kgf	or W2 = 23360 Kgf
Path 3-3 = 35198	Kgf, must exceed W = 12423 Kgf	or W3 = 23360 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 41 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.4459 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 265.4459 mm.

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Input, Nozzle Desc: K2-2"

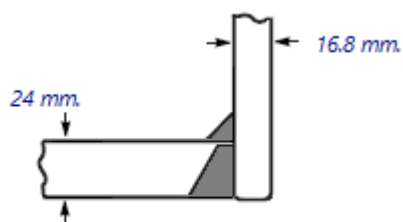
From: 20

Pressure for Reinforcement Calculations	P	41.005	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1231.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		500.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	240.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 41 \cdot 616 / (138 \cdot 1 - 0.6 \cdot 41)$
 $= 18.6520$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([41 / (115 \cdot 1)] - 1))$
 $= 0.9584$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 130.3280 mm.
 Parallel to Vessel Wall Rn+tn+t 65.1640 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.4100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

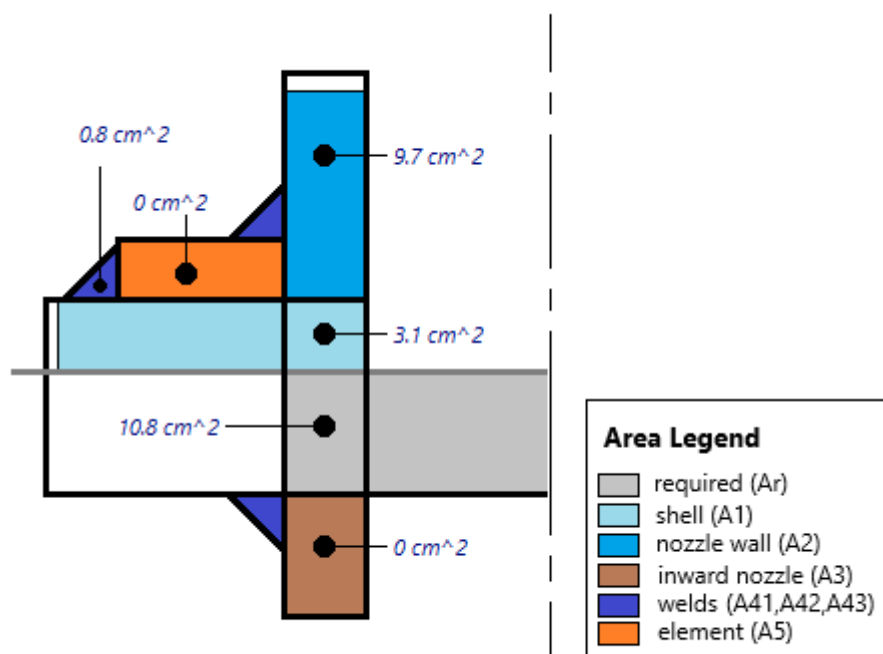


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10.828	NA	NA
Area in Shell	A1	3.142	NA	NA
Area in Nozzle Wall	A2	9.725	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA

Area in Welds	A41+A42+A43	0.833	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	13.700	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (52.8 \cdot 18.7 \cdot 1 + 2 \cdot 15.8 \cdot 18.7 \cdot 1 \cdot (1 - 0.83))$
 $= 10.828 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.5(1 \cdot 23 - 1 \cdot 18.7) - 2 \cdot 15.8$
 $(1 \cdot 23 - 1 \cdot 18.7) \cdot (1 - 0.83)$
 $= 3.142 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 39.4)(15.8 - 0.96)0.83$
 $= 9.725 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$
 $= 0.833 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 1.9584 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 19.6520 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 19.6520 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 5.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(19.7, 2.5)]$
 $= 5.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.96, 5.8)$
 $= 5.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = $0.7 \cdot Wo \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (10.8 - 3.14 + 2 \cdot 15.8 \cdot 0.83 \cdot \\
 &\quad (1 \cdot 23 - 18.7)) \cdot 138) \\
 &= 12423.09 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v \\
 &= (9.72 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138 \\
 &= 14856.99 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\
 &= (9.72 + 0 + 0.83 + (6.04)) \cdot 138 \\
 &= 23360.25 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (9.72 + 0 + 0.83 + 0 + (6.04)) \cdot 138 \\
 &= 23360.25 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115 \\
 &= 7611. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115 \\
 &= 13936. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{g n v i} \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 84.3 \cdot (20 - 0) \cdot 0.74 \cdot 138 \\
 &= 27587. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7611 + 13936) = 21547 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (7611 + 0 + 27587 + 0) = 35198 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (7611 + 27587 + 0) = 35198 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 12423 Kgf	or W1 = 14856 Kgf
Path 2-2 = 35198	Kgf, must exceed W = 12423 Kgf	or W2 = 23360 Kgf
Path 3-3 = 35198	Kgf, must exceed W = 12423 Kgf	or W3 = 23360 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 41 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.4459 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 265.4459 mm.

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Input, Nozzle Desc: K3-2"

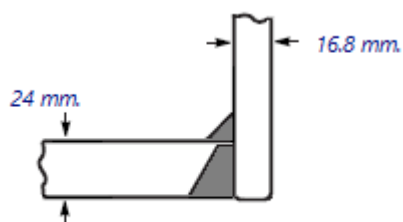
From: 20

Pressure for Reinforcement Calculations	P	41.005	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1231.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		650.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	240.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 41 \cdot 616 / (138 \cdot 1 - 0.6 \cdot 41)$
 $= 18.6520$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([41 / (115 \cdot 1)] - 1))$
 $= 0.9584$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 130.3280 mm.
 Parallel to Vessel Wall Rn+tn+t 65.1640 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.4100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

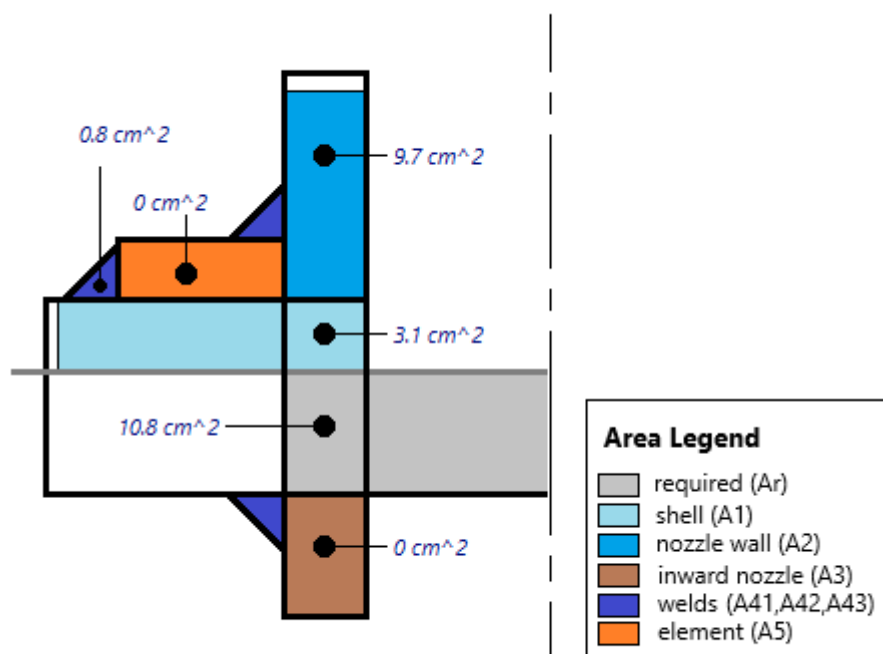


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10.828	NA	NA
Area in Shell	A1	3.142	NA	NA
Area in Nozzle Wall	A2	9.725	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	0.833	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	13.700	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (52.8 \cdot 18.7 \cdot 1 + 2 \cdot 15.8 \cdot 18.7 \cdot 1 \cdot (1 - 0.83))$$

$$= 10.828 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 77.5(1 \cdot 23 - 1 \cdot 18.7) - 2 \cdot 15.8$$

$$(1 \cdot 23 - 1 \cdot 18.7) \cdot (1 - 0.83)$$

$$= 3.142 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 39.4)(15.8 - 0.96)0.83$$

$$= 9.725 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$$

$$= 0.833 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9584 mm.
Wall Thickness per UG16(b),	tr16b = 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 19.6520 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 19.6520 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm.
Wall Thickness per table UG-45	tb3 = 5.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[5.8, \max(19.7, 2.5)]$$

$$= 5.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.96, 5.8)$$

$$= 5.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (10.8 - 3.14 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 23 - 18.7)) \cdot 138) \\
 &= 12423.09 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v \\
 &= (9.72 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138 \\
 &= 14856.99 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\
 &= (9.72 + 0 + 0.83 + (6.04)) \cdot 138 \\
 &= 23360.25 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (9.72 + 0 + 0.83 + 0 + (6.04)) \cdot 138 \\
 &= 23360.25 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115 \\
 &= 7611. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115 \\
 &= 13936. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{g n v i} \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 84.3 \cdot (20 - 0) \cdot 0.74 \cdot 138 \\
 &= 27587. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7611 + 13936) = 21547 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (7611 + 0 + 27587 + 0) = 35198 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (7611 + 27587 + 0) = 35198 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 12423 Kgf	or W1 = 14856 Kgf
Path 2-2 = 35198	Kgf, must exceed W = 12423 Kgf	or W2 = 23360 Kgf
Path 3-3 = 35198	Kgf, must exceed W = 12423 Kgf	or W3 = 23360 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 41 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.4459 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 265.4459 mm.

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Input, Nozzle Desc: K4-2"

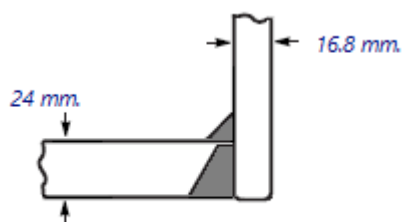
From: 20

Pressure for Reinforcement Calculations	P	41.005	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1231.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		300.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	115.00	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	240.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 2.3	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: K4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 41 \cdot 616 / (138 \cdot 1 - 0.6 \cdot 41)$
 $= 18.6520$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([41 / (115 \cdot 1)] - 1))$
 $= 0.9584$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 130.3280 mm.
 Parallel to Vessel Wall Rn+tn+t 65.1640 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.4100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 115 / 138)$
 $= 0.833$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.83, 1)$
 $= 0.833$

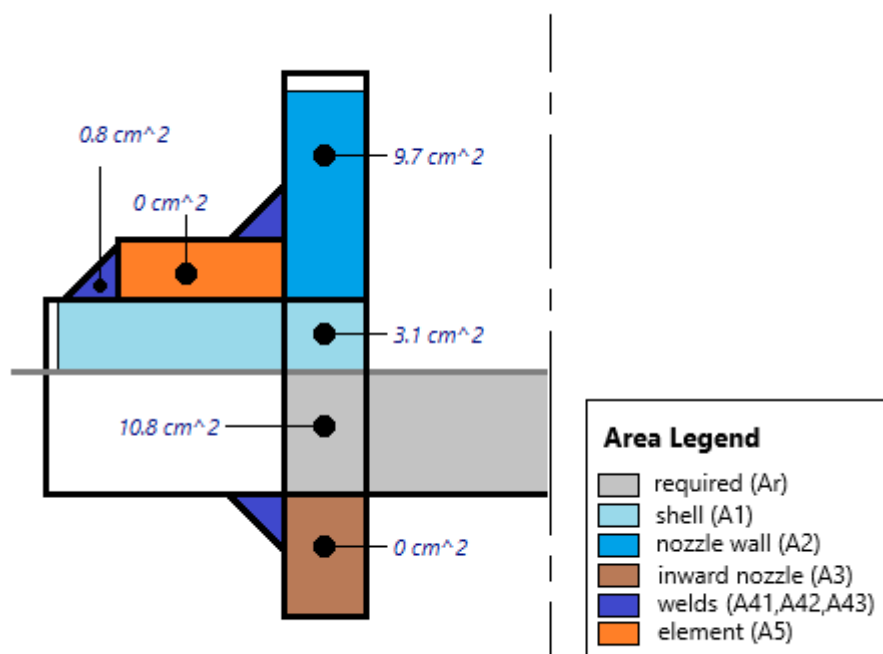


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10.828	NA	NA
Area in Shell	A1	3.142	NA	NA
Area in Nozzle Wall	A2	9.725	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA



Area in Welds	A41+A42+A43	0.833	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	13.700	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (52.8 \cdot 18.7 \cdot 1 + 2 \cdot 15.8 \cdot 18.7 \cdot 1 \cdot (1 - 0.83))$$

$$= 10.828 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 77.5(1 \cdot 23 - 1 \cdot 18.7) - 2 \cdot 15.8$$

$$(1 \cdot 23 - 1 \cdot 18.7) \cdot (1 - 0.83)$$

$$= 3.142 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 39.4)(15.8 - 0.96)0.83$$

$$= 9.725 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.83 + (0)^2 \cdot 0.83$$

$$= 0.833 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9584 mm.
Wall Thickness per UG16(b),	tr16b = 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 19.6520 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 19.6520 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm.
Wall Thickness per table UG-45	tb3 = 5.8000 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[5.8, \max(19.7, 2.5)]$$

$$= 5.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.96, 5.8)$$

$$= 5.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:



$$\begin{aligned}
 &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\
 &= \max(0, (10.8 - 3.14 + 2 \cdot 15.8 \cdot 0.83 \cdot (1 \cdot 23 - 18.7)) \cdot 138) \\
 &= 12423.09 \text{ Kgf}
 \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned}
 &= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v \\
 &= (9.72 + 0 + 0.83 - 0 \cdot 0.83) \cdot 138 \\
 &= 14856.99 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\
 &= (9.72 + 0 + 0.83 + (6.04)) \cdot 138 \\
 &= 23360.25 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (9.72 + 0 + 0.83 + 0 + (6.04)) \cdot 138 \\
 &= 23360.25 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 84.3 \cdot 10 \cdot 0.49 \cdot 115 \\
 &= 7611. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 34.3) \cdot (16.8 - 1) \cdot 0.7 \cdot 115 \\
 &= 13936. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{g n v i} \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 84.3 \cdot (20 - 0) \cdot 0.74 \cdot 138 \\
 &= 27587. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (7611 + 13936) = 21547 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\
 &= (7611 + 0 + 27587 + 0) = 35198 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\
 &= (7611 + 27587 + 0) = 35198 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 21547	Kgf, must exceed W = 12423 Kgf	or W1 = 14856 Kgf
Path 2-2 = 35198	Kgf, must exceed W = 12423 Kgf	or W2 = 23360 Kgf
Path 3-3 = 35198	Kgf, must exceed W = 12423 Kgf	or W3 = 23360 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 41 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.4459 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 265.4459 mm.

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Input, Nozzle Desc: M1-24"

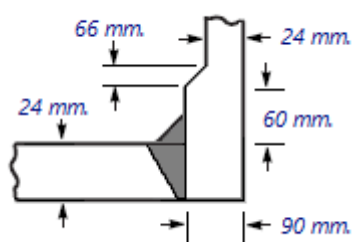
From: 20

Pressure for Reinforcement Calculations	P	41.005	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1231.00	mm.
Shell Finished (Minimum) Thickness	t	24.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		900.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	24.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	60.0000	mm.
Height of Beveled Transition	L'	66.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm.
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	430.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: M1-24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.945 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 41 \cdot 616 / (138 \cdot 1 - 0.6 \cdot 41)$
 $= 18.6520$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 41 \cdot 306 / (138 \cdot 1 - 0.6 \cdot 41)$
 $= 9.2520$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 66 / 0.5773$
 $= 114.3154$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

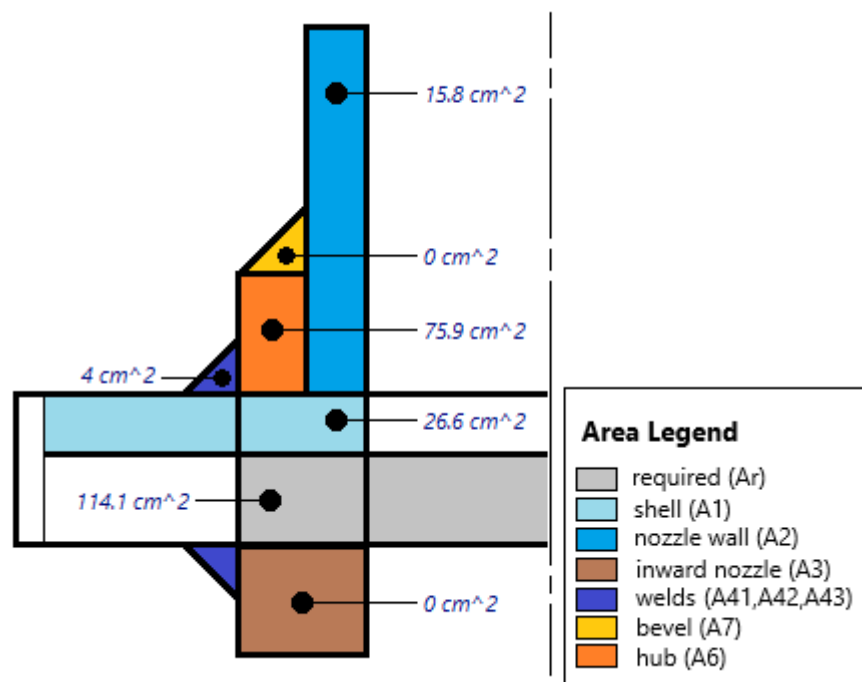
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1223.2001	mm.
Parallel to Vessel Wall, opening length	d	611.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	57.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	114.076	NA	NA
Area in Shell	A1	26.592	NA	NA
Area in Nozzle Wall	A2	15.810	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	75.899	NA	NA
TOTAL AREA AVAILABLE	Atot	122.301	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (612 \cdot 18.7 \cdot 1 + 2 \cdot 89 \cdot 18.7 \cdot 1 \cdot (1 - 1))$$

$$= 114.076 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 612 (1 \cdot 23 - 1 \cdot 18.7) - 2 \cdot 89 (1 \cdot 23 - 1 \cdot 18.7) \cdot (1 - 1)$$

$$= 26.592 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 57.5) (23 - 9.25) 1$$

$$= 15.810 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 20^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(57.5, 430, 60)) \cdot (90 - 24) \cdot 1$$

$$= 75.899 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 10.2520 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= t_a$
 $= 10.2520 \text{ mm.}$

Available Nozzle Neck Thickness = 24.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 24$, $t_r = 9.25$, $c = 1 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= t_r \cdot E* / (t_g - c) = 0.4$, Temp. Reduction = 50 °C

Min Metal Temp. w/o impact per UCS-66, Curve A $18 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-33 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 24$, $t_r = 18.7$, $c = 1 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= t_r \cdot E* / (t_g - c) = 0.81$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B $-2 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-13 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-13 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) $-29 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $41.00/51.00 = 0.804$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (H_{ub} \text{ Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 66/0.5773$
 $= 114.3154 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) $D1 \quad 917.4000 \text{ mm.}$
 Parallel to Vessel Wall, opening length $d \quad 458.7000 \text{ mm.}$
 Normal to Vessel Wall (Thickness Limit), no pad $T_{lnp} \quad 57.5000 \text{ mm.}$

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	76.055	76.055
Area Available in Shell	A1	13.296	13.296
Area Available in Nozzle Wall	A2	15.810	15.810
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	4.000	4.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	75.899	75.899
TOTAL AREA AVAILABLE	Atot	109.005	109.005

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (114 - 26.6 + 2 \cdot 89 \cdot 1 \cdot (1 \cdot 23 - 18.7)) \cdot 138)$
 $= 133996.28 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (91.7 + 4 - 0 \cdot 1) \cdot 138$
 $= 134679.20 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (91.7 + 0 + 4 + (40.9)) \cdot 138$
 $= 192288.23 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (91.7 + 0 + 4 + 0 + (40.9)) \cdot 138$
 $= 192288.23 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 790 \cdot 20 \cdot 0.49 \cdot 138$
 $= 171039. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 350) \cdot (90 - 1) \cdot 0.7 \cdot 138$
 $= 964763. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 790 \cdot (20 - 0) \cdot 0.74 \cdot 138$
 $= 258309. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (171039 + 964763) = 1135802 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (171039 + 0 + 258309 + 0) = 429348 \text{ Kgf}$

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (171039 + 258309 + 0) = 429348 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 1135802 Kgf, must exceed W = 133996 Kgf or W1 = 134679 Kgf
Path 2-2 = 429347 Kgf, must exceed W = 133996 Kgf or W2 = 192288 Kgf
Path 3-3 = 429347 Kgf, must exceed W = 133996 Kgf or W3 = 192288 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 41 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 143.2996 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 597.2996 mm.

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 395.53 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 611.60 mm.

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Input, Nozzle Desc: N2-4"

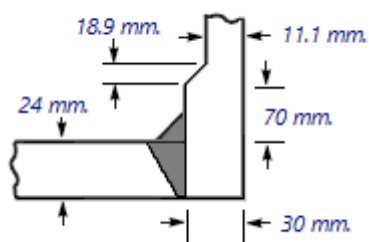
From: 60

Pressure for Reinforcement Calculations	P	41.005	bars
Temperature for Internal Pressure	Temp	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1231.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	18.8700	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	280.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		19.5000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Overall MAWP.



Reinforcement CALCULATION, Description: N2-4"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 4.500 in.
 Actual Thickness Used in Calculation 0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 41 \cdot 1233 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 41)$
 $= 16.5110$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 41 \cdot 57.1 / (138 \cdot 1 + 0.4 \cdot 41)$
 $= 1.6783$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 18.9 / 0.5773$
 $= 32.6921$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

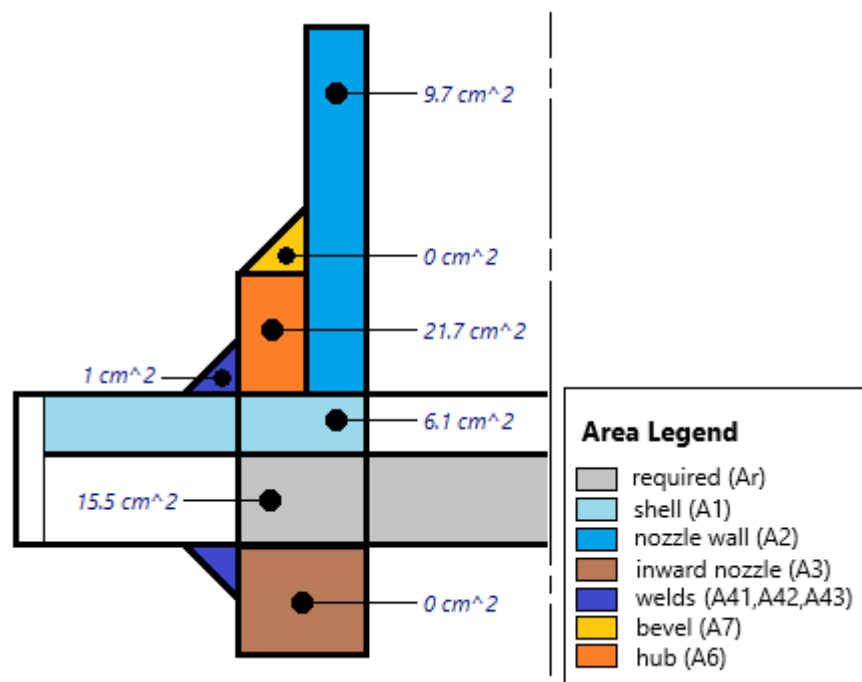
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	188.0992	mm.
Parallel to Vessel Wall, opening length	d	94.0496	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	57.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	15.529	NA	NA
Area in Shell	A1	6.103	NA	NA
Area in Nozzle Wall	A2	9.714	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	21.706	NA	NA
TOTAL AREA AVAILABLE	Atot	38.522	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (94 \cdot 16.5 \cdot 1 + 2 \cdot 29 \cdot 16.5 \cdot 1 \cdot (1 - 1))$
 $= 15.529 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 94(1 \cdot 23 - 1 \cdot 16.5) - 2 \cdot 29(1 \cdot 23 - 1 \cdot 16.5) \cdot (1 - 1)$
 $= 6.103 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 57.5)(10.1 - 1.68)1$
 $= 9.714 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2}$
 $= (2 \cdot \min(57.5, 280, 70)) \cdot (30 - 11.1) \cdot 1$
 $= 21.706 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 2.6783 \text{ mm.}$
 Wall Thickness per UG16(b), $tr_{16b} = 2.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb_1 = 19.3344 \text{ mm.}$
 Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 19.3344 \text{ mm.}$
 Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 2.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb_3 = 6.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[6.26, \max(19.3, 2.5)]$
 $= 6.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(2.68, 6.26)$
 $= 6.2578 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	46.9,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	298.1 N./mm ²	Passed
Occasional	:	8.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	22.5,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.73$, $t_r = 1.68$, $c = 1 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.19$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-104 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, $t_g = 24$, $t_r = 16.5$, $c = 1 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.72$, Temp. Reduction = $16 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-18 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -18 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $41.00/102.00 = 0.402$

Weld Size Calculations, Description: N2-4"

Intermediate Calc. for nozzle/shell Welds	T_{min}	19.0000 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

$$\begin{aligned}\text{Weld Load [W]:} \\ &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (15.5 - 6.1 + 2 \cdot 29 \cdot 1 \cdot (1 \cdot 23 - 16.5)) \cdot 138) \\ &= 18560.03 \text{ Kgf}\end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

$$\begin{aligned}\text{Weld Load [W1]:} \\ &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (31.4 + 1 - 0 \cdot 1) \cdot 138 \\ &= 45619.96 \text{ Kgf}\end{aligned}$$

$$\begin{aligned}\text{Weld Load [W2]:} \\ &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (31.4 + 0 + 1 + (13.3)) \cdot 138 \\ &= 64391.43 \text{ Kgf}\end{aligned}$$

$$\begin{aligned}\text{Weld Load [W3]:} \\ &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (31.4 + 0 + 1 + 0 + (13.3)) \cdot 138 \\ &= 64391.43 \text{ Kgf}\end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

$$\begin{aligned}\text{Shear, Outward Nozzle Weld [Sonw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 114 \cdot 10 \cdot 0.49 \cdot 138 \\ &= 12380. \text{ Kgf}\end{aligned}$$

$$\begin{aligned}\text{Shear, Nozzle Wall [Snw]:} \\ &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 52.1) \cdot (30 - 1) \cdot 0.7 \cdot 138 \\ &= 46743. \text{ Kgf}\end{aligned}$$

$$\begin{aligned}\text{Tension, Shell Groove Weld [Tngw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 114 \cdot (19.5 - 0) \cdot 0.74 \cdot 138 \\ &= 36457. \text{ Kgf}\end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

$$\begin{aligned}\text{Failure Path [Path 1-1]:} \\ &= (\text{SONW} + \text{SNW}) = (12380 + 46743) = 59123 \text{ Kgf}\end{aligned}$$

$$\begin{aligned}\text{Failure Path [Path 2-2]:} \\ &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (12380 + 0 + 36457 + 0) = 48837 \text{ Kgf}\end{aligned}$$

$$\begin{aligned}\text{Failure Path [Path 3-3]:} \\ &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (12380 + 36457 + 0) = 48837 \text{ Kgf}\end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 59123	Kgf, must exceed W = 18560 Kgf	or W1 = 45619 Kgf
Path 2-2 = 48836	Kgf, must exceed W = 18560 Kgf	or W2 = 64391 Kgf
Path 3-3 = 48836	Kgf, must exceed W = 18560 Kgf	or W3 = 64391 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 41 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.



The Drop for this Nozzle is : 1.4498 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 305.4498 mm.

Input Echo, WRC107/537 Item 1, Description: N2-4" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	2215.800	mm.
Vessel Thickness	Tv	24.000	mm.
Design Temperature	T1	130.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.0000	mm.
Nozzle Diameter	Dn	152.050	mm.
Nozzle Thickness	Tn	30.000	mm.
Nozzle Material		SA-266 2	
Nozzle UNS Number		K03506	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	41.005	bars
Include Pressure Thrust		No	

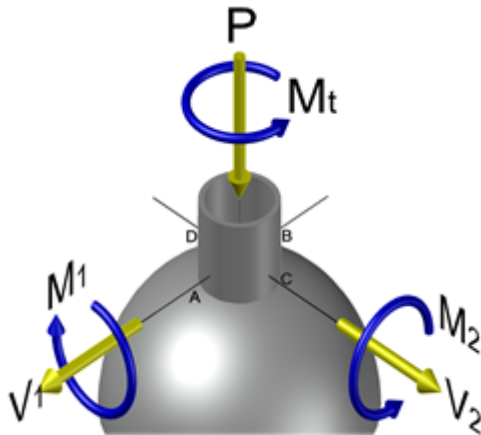
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	917.8	Kgf
Longitudinal Shear (SUS)	(V1) V1	662.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	662.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	203.9	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	203.9	Kg-m.
Torsional Moment (SUS)	Mt	234.5	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 152.05 + 2 \cdot 1.65 \cdot \sqrt{(1120.4 (24.0 - 1.0))}$$

$$= 681.792 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	917.8	Kgf
Circumferential Shear	(VC) V2	662.9	Kgf
Longitudinal Shear	(VL) V1	662.9	Kgf
Circumferential Moment	(MC) M1	203.9	Kg-m.
Longitudinal Moment	(ML) M2	203.9	Kg-m.
Torsional Moment	MT	234.5	Kg-m.

Dimensionless Param: U = 0.47 TAU = 5.00 (2.12) RHO = 0.79

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.07641	(A,B,C,D)
$M(x) / P$	SP 1	0.09409	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.12960	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.24614	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.12960	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.24614	(A,B,C,D)
$N(y) * T / P$	SP 1	0.13459	(A,B,C,D)
$M(y) / P$	SP 1	0.03297	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.08947	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.10000	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.08947	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.10000	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3	-1.3
Rad. Bend.	P	-9.6	9.6	-9.6	9.6	-9.6	9.6	-9.6	9.6
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-3.1	-3.1	3.1	3.1
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-34.8	34.8	34.8	-34.8
Rad. Memb.	ML	-3.1	-3.1	3.1	3.1	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-34.8	34.8	34.8	-34.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-48.7	40.0	26.9	-23.4	-48.7	40.0	26.9	-23.4



Tang. Memb. P	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3	-2.3
Tang. Bend. P	-3.4	3.4	-3.4	3.4	-3.4	3.4	-3.4	3.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-2.1	-2.1	2.1	2.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-14.1	14.1	14.1	-14.1
Tang. Memb. ML	-2.1	-2.1	2.1	2.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-14.1	14.1	14.1	-14.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-21.9	13.1	10.6	-10.9	-21.9	13.1	10.6	-10.9
Shear VC	1.2	1.2	-1.2	-1.2	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear MT	2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
Tot. Shear	3.9	3.9	1.6	1.6	1.6	1.6	3.9	3.9
Str. Int.	49.3	40.6	27.1	23.6	48.8	40.1	27.8	24.6

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9
Rad. Pl (SUS)		-4.4	-4.4	1.8	1.8	-4.4	-4.4	1.8	1.8
Rad. Q (SUS)		-44.4	44.4	25.2	-25.2	-44.4	44.4	25.2	-25.2
Long. Pm (SUS)		98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9
Long. Pl (SUS)		-4.4	-4.4	-0.2	-0.2	-4.4	-4.4	-0.2	-0.2
Long. Q (SUS)		-17.5	17.5	10.8	-10.8	-17.5	17.5	10.8	-10.8
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.2	1.2	-1.2	-1.2	-1.2	-1.2	1.2	1.2
Shear Q (SUS)		2.8	2.8	2.8	2.8	2.8	2.8	2.8	2.8
Pm (SUS)		98.9	98.9	98.9	98.9	98.9	98.9	98.9	98.9
Pm+Pl (SUS)		95.7	95.7	101.2	101.2	95.7	95.7	101.2	101.2
Pm+Pl+Q (Total)		77.6	139.5	125.9	88.1	77.1	139.0	126.7	89.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	98.88	138.00	Passed
Pm+Pl (SUS)	101.19	207.00	Passed
Pm+Pl+Q (TOTAL)	139.47	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: